

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T10:10:18

Source: data/json/noise_experiments/v2/noise2i_match8_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_match8/noise2i_match8_game_001_neutral_rep01.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_match8
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 1
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 0
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 1
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$3.5; payoffs Agent_1=\$5.5, Agent_2=\$5.5;

noise/visible: sent visible=\$2.94, received visible=\$8.82, returned visible=\$3.02

Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 5.5,

"Agent_2": 5.5

}

Visible balances after round: {

"Agent_1": 5.02,

"Agent_2": 5.32

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:06:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

I'll start with a moderate level of trust to see how the receiver responds.

Since this is a multi-round game, I want to signal cooperation while being somewhat cautious in the first round. Sending a moderate amount allows us both to benefit from the 3x multiplier while I assess the receiver's trustworthiness.

I'll send \$3, which becomes \$9 for the receiver. This gives them a good opportunity to demonstrate trustworthiness by returning a fair amount, and we can build mutual trust from there.

```
{'send': 3}
```

Reasoning:

This is the first round of a trust game where I'm playing as the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- There's communication noise, so amounts may differ from what was actually sent/returned

This is a classic trust game. The strategic considerations are:

1. If I send nothing (\$0), I keep \$5, the receiver gets \$0. Total payoff: me=\$5, them=\$0, combined=\$5
2. If I send everything (\$5), the receiver gets \$15, and then decides how much to return. If they return half (\$7.50), I'd end up with \$7.50 and they'd keep \$7.50. Total combined=\$15
3. The efficient outcome is to send as much as possible since it gets multiplied by 3x, maximizing total welfare. The question is whether I can trust the receiver to share fairly.

First round considerations:

- No reputation or history yet
- The receiver doesn't know what kind of player I am
- There's communication noise which adds uncertainty
- This is a multi-round game, so building trust could be valuable

Common strategies in trust games:

1. Send nothing (zero trust)
2. Send half (\$2.50) as a moderate trust signal
3. Send everything (\$5) as a strong trust signal

Since this is round 1 and there are multiple rounds, I should consider:

- Sending some amount shows willingness to cooperate
- Sending too much risks being exploited
- Sending a moderate amount (like \$2-3) could be a good starting point to test the waters
- The noise factor adds uncertainty, so the receiver might see a slightly different amount

I'll send \$3 as a moderate trust signal. This shows I'm willing to cooperate but not completely naive. The receiver will get \$9, and hopefully will return a fair amount (ideally \$4-5 would give us both a good payoff).

...

```
Usage: {  
  "input_tokens": 365,  
  "output_tokens": 620,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:06:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
  },
```

```

    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.94, that is 58.8% of their total endowment. You received \$8.82. How much do you return to The sender? (0-8.82)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.94 (58.8% of their endowment)
- This created \$8.82 for me to work with
- The sender currently has only \$2.06 left

To reciprocate their trust and ensure mutual benefit, I should return a fair amount. The sender took a risk, and both of us should gain from cooperation.

If I return \$3.50:

- Sender ends with: \$2.06 + \$3.50 = \$5.56 (modest gain)
- I keep: \$8.82 - \$3.50 = \$5.32 (modest gain)

This seems like a fair split that rewards the sender's trust while we both benefit.

```
{'return': 3.50}
```

Reasoning:

Let me think about this game theory situation carefully.

As the RECEIVER:

- The sender sent \$2.94 out of their \$5 endowment (58.8%)
- This amount was multiplied by 3, so I received $\$2.94 \times 3 = \8.82
- I need to decide how much to return (between \$0 and \$8.82)

Key considerations:

1. The sender showed significant trust by sending nearly 60% of their endowment
2. If I return nothing, I keep \$8.82 and the sender gets only $\$5 - \$2.94 = \$2.06$
3. If I return a fair amount, I should consider:
 - The sender's initial position: they had \$5
 - They sent \$2.94, keeping \$2.06
 - I received \$8.82
 - Total money created: $\$2.06$ (sender kept) + $\$8.82$ (I received) = $\$10.88$

A fair approach might be to:

- Split the gains from cooperation
- The sender started with \$5, gave up \$2.94 to create \$8.82
- Net gain from the transaction: $\$8.82 - \$2.94 = \$5.88$
- If we split this gain equally: $\$5.88 / 2 = \2.94 each

So the sender should end up with: $\$2.06$ (what they kept) + $\$2.94$ (their share of gains) = $\$5.00$

This means I should return: \$2.94

Alternatively, I could think about it as:

- The sender took a risk and sent a substantial amount
- A reasonable return might be to ensure the sender doesn't lose money
- Or to ensure both parties benefit
- Testing different return amounts: returning \$3.50 would give them \$5.56 while I keep \$5.32, which rewards their trust while still letting me benefit from the cooperation....

Usage: {

```
"input_tokens": 394,  
"output_tokens": 899,  
"reasoning_tokens": 0
```

}

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

```
dyad_1: Agent_2 sent $3; received $9; Agent_1 returned $0.75; payoffs Agent_2=$2.75,  
Agent_1=$8.25; noise/visible: sent visible=$0.65, received visible=$1.95, returned visible=$1.12  
Action validation: {"returned": {"amount": 0.75, "clamped": false, "max": 9.0, "min": 0, "raw":  
0.75}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
```

Balances after round: {

```
"Agent_1": 13.75,  
"Agent_2": 8.25
```

}

Visible balances after round: {

```
"Agent_1": 6.22,  
"Agent_2": 8.440000000000001
```

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:06:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_1",  
"move_index": 0,
```

```

"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.94, that is 58.8% of their total endowment. You received \$8.82. How much do you return to The sender? (0-8.82)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.94 (58.8% of their endowment)
- This created \$8.82 for me to work with
- The sender currently has only \$2.06 left

To reciprocate their trust and ensure mutual benefit, I should return a fair amount. The sender took a risk, and both of us should gain from cooperation.

If I return \$3.50:

- Sender ends with: $\$2.06 + \$3.50 = \$5.56$ (modest gain)
- I keep: $\$8.82 - \$3.50 = \$5.32$ (modest gain)

This seems like a fair split that rewards the sender's trust while we both benefit.

```
{'return': 3.50}
```

reasoning:

Let me think about this game theory situation carefully.

As the RECEIVER:

- The sender sent \$2.94 out of their \$5 endowment (58.8%)
- This amount was multiplied by 3, so I received $\$2.94 \times 3 = \8.82
- I need to decide how much to return (between \$0 and \$8.82)

Key considerations:

1. The sender showed significant trust by sending nearly 60% of their endowment
2. If I return nothing, I keep \$8.82 and the sender gets only $\$5 - \$2.94 = \$2.06$
3. If I return a fair amount, I should consider:
 - The sender's initial position: they had \$5
 - They sent \$2.94, keeping \$2.06
 - I received \$8.82
 - Total money created: $\$2.06$ (sender kept) + $\$8.82$ (I received) = $\$10.88$

A fair approach might be to:

- Split the gains from cooperation
- The sender started with \$5, gave up \$2.94 to create \$8.82
- Net gain from the transaction: $\$8.82 - \$2.94 = \$5.88$
- If we split this gain equally: $\$5.88 / 2 = \2.94 each

So the sender should end up with: $\$2.06$ (what they kept) + $\$2.94$ (their share of gains) = $\$5.00$

This means I should return: \$2.94

Alternatively, I could think about it as:

- The sender took a risk and sent a substantial amount
- A reasonable return might be to ensure the sender doesn't lose money
- Or to ensure both parties benefit
- Testing different return amounts: returning \$3.50 would give them \$5.56 while I keep \$5.32, which rewards their trust while still letting me benefit from the cooperation....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 899,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.32.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the situation:

In Round 1, the sender showed trust by sending 58.8% of their endowment. I reciprocated fairly by returning a meaningful amount, and we both benefited.

Now that I'm the sender, I want to continue this cooperative approach. Sending a similar proportion demonstrates that I'm willing to trust the receiver just as the previous sender trusted me.

If I send \$3.00:

- The receiver will get \$9.00
- If they reciprocate fairly (as I did), we should both benefit
- This maintains the cooperative pattern

```
{'send': 3.00}
```

Reasoning:

Now I'm the sender in Round 2. Let me think about this strategically:

Round 1 recap:

- I was the receiver
- Sender sent me \$2.94 (58.8% of their endowment)
- I received \$8.82
- I returned \$3.50
- I earned \$5.32 total

Now I'm the sender:

- I get \$5 this round
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

Strategy considerations:

1. The previous sender showed trust by sending 58.8%
2. I reciprocated fairly by returning \$3.50, which gave them a modest gain
3. This established a cooperative pattern
4. I should probably continue this cooperative approach

If I send around the same proportion (58.8%), that would be about \$2.94.

But let me think about what makes sense:

If I send \$3:

- Receiver gets \$9
- If they return proportionally (like I did, roughly 40% of what they received), they'd return about \$3.60
- I'd end with: $\$2 + \$3.60 = \$5.60$
- They'd end with: $\$9 - \$3.60 = \$5.40$

If I send \$2.50:

- Receiver gets \$7.50
- If they return similarly (40%), they might return about \$3
- I'd end with: $\$2.50 + \$3 = \$5.50$
- They'd end with: $\$7.50 - \$3 = \$4.50$

If I send \$2.94 (matching what was sent to me):

- Receiver gets \$8.82

...

```
Usage: {  
  "input_tokens": 622,  
  "output_tokens": 684,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:07:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,
```

```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderate level of trust to see how the receiver responds.

Since this is a multi-round game, I want to signal cooperation while being somewhat cautious in the first round. Sending a moderate amount allows us both to benefit from the 3x multiplier while I assess the receiver's trustworthiness.

I'll send \$3, which becomes \$9 for the receiver. This gives them a good opportunity to demonstrate trustworthiness by returning a fair amount, and we can build mutual trust from there.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm playing as the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- There's communication noise, so amounts may differ from what was actually sent/returned

This is a classic trust game. The strategic considerations are:

1. If I send nothing (\$0), I keep \$5, the receiver gets \$0. Total payoff: me=\$5, them=\$0, combined=\$5

2. If I send everything (\$5), the receiver gets \$15, and then decides how much to return. If they return half (\$7.50), I'd end up with \$7.50 and they'd keep \$7.50. Total combined = \$15
3. The efficient outcome is to send as much as possible since it gets multiplied by 3x, maximizing total welfare. The question is whether I can trust the receiver to share fairly.

First round considerations:

- No reputation or history yet
- The receiver doesn't know what kind of player I am
- There's communication noise which adds uncertainty
- This is a multi-round game, so building trust could be valuable

Common strategies in trust games:

1. Send nothing (zero trust)
2. Send half (\$2.50) as a moderate trust signal
3. Send everything (\$5) as a strong trust signal

Since this is round 1 and there are multiple rounds, I should consider:

- Sending some amount shows willingness to cooperate
- Sending too much risks being exploited
- Sending a moderate amount (like \$2-3) could be a good starting point to test the waters
- The noise factor adds uncertainty, so the receiver might see a slightly different amount

I'll send \$3 as a moderate trust signal. This shows I'm willing to cooperate but not completely naive. The receiver will get \$9, and hopefully will return a fair amount (ideally \$4-5 would give us both a good payoff).

...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 620,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$5.02.

This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Assistant output:

Looking at this situation:

- The sender sent me \$0.65, which became \$1.95 after the 3x multiplier
- This is a relatively small amount, suggesting caution on their part
- In the previous round, I received a good return when I sent \$3

I want to demonstrate trustworthiness and encourage cooperation in future rounds. A fair return would give the sender back more than they initially sent while still benefiting from the multiplier effect.

I'll return \$0.75, which:

- Gives the sender a small profit (\$0.75 > \$0.65 sent)
- Shows I'm reciprocating fairly
- Leaves me with \$1.20 from this exchange
- May encourage more trust in future rounds

```
{'return': 0.75}
```

Reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3.
2. My total earnings across all rounds are \$5.02
3. This means in Round 1: I sent \$3, so I kept \$2 initially. My total earnings are \$5.02, which means I must have received back \$3.02 from the receiver.
4. So the receiver returned \$3.02 out of the \$9 they received (I sent $3 \times 3 = 9$)

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0.65
- I received \$1.95 (0.65×3)
- The note mentions there may be communication noise, so amounts may differ slightly

The previous receiver returned about 33.6% of what they received (\$3.02 out of \$9). This seems like a reasonable reciprocal behavior.

Since the sender only sent me \$0.65 (which is quite low), they are being cautious. However, I should still demonstrate some level of trustworthiness to encourage cooperation in future rounds.

A fair return might be around 1/3 to 1/2 of what I received, which would be \$0.65 to \$0.98.

Let me return about \$0.75, which is roughly 38% of what I received. This:

- Shows I'm trustworthy and willing to reciprocate
- Leaves me with about \$1.20 from this round
- Gives the sender back more than they sent me originally ($\$0.75 > \0.65)
- May encourage them to send more in future rounds

Actually, let me think about this more carefully. The sender sent \$0.65 and it became \$1.95. .

```
Usage: {
  "input_tokens": 561,
  "output_tokens": 663,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$0; payoffs Agent_1=\$2, Agent_2=\$9;
noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 15.75,
"Agent_2": 17.25

}

Visible balances after round: {

"Agent_1": 8.219999999999999,
"Agent_2": 8.440000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:07:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 }
},

```

    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderate level of trust to see how the receiver responds.

Since this is a multi-round game, I want to signal cooperation while being somewhat cautious in the first round. Sending a moderate amount allows us both to benefit from the 3x multiplier while I assess the receiver's trustworthiness.

I'll send \$3, which becomes \$9 for the receiver. This gives them a good opportunity to demonstrate trustworthiness by returning a fair amount, and we can build mutual trust from there.

```
{'send': 3}
```

reasoning:

This is the first round of a trust game where I'm playing as the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- There's communication noise, so amounts may differ from what was actually sent/returned

This is a classic trust game. The strategic considerations are:

1. If I send nothing (\$0), I keep \$5, the receiver gets \$0. Total payoff: me=\$5, them=\$0, combined=\$5
2. If I send everything (\$5), the receiver gets \$15, and then decides how much to return. If they return half (\$7.50), I'd end up with \$7.50 and they'd keep \$7.50. Total combined=\$15
3. The efficient outcome is to send as much as possible since it gets multiplied by 3x, maximizing total welfare. The question is whether I can trust the receiver to share fairly.

First round considerations:

- No reputation or history yet
- The receiver doesn't know what kind of player I am
- There's communication noise which adds uncertainty
- This is a multi-round game, so building trust could be valuable

Common strategies in trust games:

1. Send nothing (zero trust)
2. Send half (\$2.50) as a moderate trust signal
3. Send everything (\$5) as a strong trust signal

Since this is round 1 and there are multiple rounds, I should consider:

- Sending some amount shows willingness to cooperate
- Sending too much risks being exploited
- Sending a moderate amount (like \$2-3) could be a good starting point to test the waters
- The noise factor adds uncertainty, so the receiver might see a slightly different amount

I'll send \$3 as a moderate trust signal. This shows I'm willing to cooperate but not completely naive. The receiver will get \$9, and hopefully will return a fair amount (ideally \$4-5 would give us both a good payoff).

...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 620,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$5.02.

This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.65, which became \$1.95 after the 3x multiplier
- This is a relatively small amount, suggesting caution on their part
- In the previous round, I received a good return when I sent \$3

I want to demonstrate trustworthiness and encourage cooperation in future rounds. A fair return would give the sender back more than they initially sent while still benefiting from the multiplier effect.

I'll return \$0.75, which:

- Gives the sender a small profit (\$0.75 > \$0.65 sent)
- Shows I'm reciprocating fairly
- Leaves me with \$1.20 from this exchange
- May encourage more trust in future rounds

```
{'return': 0.75}
```

reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3.
2. My total earnings across all rounds are \$5.02
3. This means in Round 1: I sent \$3, so I kept \$2 initially. My total earnings are \$5.02, which means I must have received back \$3.02 from the receiver.
4. So the receiver returned \$3.02 out of the \$9 they received (I sent $3 \times 3 = 9$)

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0.65
- I received \$1.95 (0.65×3)
- The note mentions there may be communication noise, so amounts may differ slightly

The previous receiver returned about 33.6% of what they received (\$3.02 out of \$9). This seems like a reasonable reciprocal behavior.

Since the sender only sent me \$0.65 (which is quite low), they are being cautious. However, I should still demonstrate some level of trustworthiness to encourage cooperation in future rounds.

A fair return might be around 1/3 to 1/2 of what I received, which would be \$0.65 to \$0.98.

Let me return about \$0.75, which is roughly 38% of what I received. This:

- Shows I'm trustworthy and willing to reciprocate
- Leaves me with about \$1.20 from this round
- Gives the sender back more than they sent me originally ($0.75 > 0.65$)
- May encourage them to send more in future rounds

Actually, let me think about this more carefully. The sender sent \$0.65 and it became \$1.95. ...

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 663,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.22.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern so far:

Round 1: I sent \$3 and received a fair return (~\$3), showing the receiver was trustworthy

Round 2: As receiver, I got \$0.65 (small amount), and I returned \$0.75 to demonstrate my trustworthiness

Since I experienced good cooperation when I sent \$3 in Round 1, and I've now shown I'm trustworthy as a receiver, I'll send \$3 again. This maintains consistency and signals continued willingness to cooperate.

```
{'send': 3}
```

Reasoning:

Let me think about what's happened so far:

Round 1: I was the SENDER

- I sent \$3
- That became \$9 for the receiver
- My earnings are stated as \$5.02 in round 2
- If I sent \$3, I had \$2 left
- So I must have received back about \$3.02 from the receiver
- That's a fair return - the receiver returned about 1/3 of what they got

Round 2: I was the RECEIVER

- The sender sent me \$0.65
- That became \$1.95 for me
- I returned \$0.75
- The sender kept $5 - 0.65 + 0.75 = 5.10$

- I kept $\$1.95 - \$0.75 = \$1.20$

Now Round 3: I'm the SENDER again

- My total earnings are \$6.22
- Let me verify: From round 1 as sender: $\$2 + \$3.02 = \$5.02$
- From round 2 as receiver: \$1.20
- Total: $\$5.02 + \$1.20 = \$6.22$?

The receiver in Round 1 was fairly generous, returning about \$3 on the \$9 they received.
In Round 2, the sender only sent \$0.65, which is quite cautious.

The game notes mention there may be communication noise - amounts may differ from what was actually sent/returned. This is interesting.

For Round 3:

- I want to continue building trust
- The receiver in Round 1 was trustworthy
- The sender in Round 2 was cautious but I showed I'm trustworthy by returning more than they sent
- I should probably send a moderate to high amount to encourage cooperation

I'll send \$3 again, which has worked well before. ...

```
Usage: {  
  "input_tokens": 783,  
  "output_tokens": 575,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:07:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.94, that is 58.8% of their total endowment. You received \$8.82. How much do you return to The sender? (0-8.82)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.94 (58.8% of their endowment)
- This created \$8.82 for me to work with
- The sender currently has only \$2.06 left

To reciprocate their trust and ensure mutual benefit, I should return a fair amount. The sender took a risk, and both of us should gain from cooperation.

If I return \$3.50:

- Sender ends with: $\$2.06 + \$3.50 = \$5.56$ (modest gain)
- I keep: $\$8.82 - \$3.50 = \$5.32$ (modest gain)

This seems like a fair split that rewards the sender's trust while we both benefit.

{'return': 3.50}

reasoning:

Let me think about this game theory situation carefully.

As the RECEIVER:

- The sender sent \$2.94 out of their \$5 endowment (58.8%)
- This amount was multiplied by 3, so I received $\$2.94 \times 3 = \8.82
- I need to decide how much to return (between \$0 and \$8.82)

Key considerations:

1. The sender showed significant trust by sending nearly 60% of their endowment
2. If I return nothing, I keep \$8.82 and the sender gets only $\$5 - \$2.94 = \$2.06$
3. If I return a fair amount, I should consider:
 - The sender's initial position: they had \$5
 - They sent \$2.94, keeping \$2.06
 - I received \$8.82
 - Total money created: $\$2.06$ (sender kept) + $\$8.82$ (I received) = $\$10.88$

A fair approach might be to:

- Split the gains from cooperation
- The sender started with \$5, gave up \$2.94 to create \$8.82
- Net gain from the transaction: $\$8.82 - \$2.94 = \$5.88$
- If we split this gain equally: $\$5.88 / 2 = \2.94 each

So the sender should end up with: $\$2.06$ (what they kept) + $\$2.94$ (their share of gains) = $\$5.00$

This means I should return: \$2.94

Alternatively, I could think about it as:

- The sender took a risk and sent a substantial amount
- A reasonable return might be to ensure the sender doesn't lose money
- Or to ensure both parties benefit
- Testing different return amounts: returning \$3.50 would give them \$5.56 while I keep \$5.32, which rewards their trust while still letting me benefit from the cooperation....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 899,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.32.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed trust by sending 58.8% of their endowment. I reciprocated fairly by returning a meaningful amount, and we both benefited.

Now that I'm the sender, I want to continue this cooperative approach. Sending a similar proportion demonstrates that I'm willing to trust the receiver just as the previous sender trusted me.

If I send \$3.00:

- The receiver will get \$9.00
- If they reciprocate fairly (as I did), we should both benefit
- This maintains the cooperative pattern

{'send': 3.00}

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically:

Round 1 recap:

- I was the receiver
- Sender sent me \$2.94 (58.8% of their endowment)
- I received \$8.82
- I returned \$3.50
- I earned \$5.32 total

Now I'm the sender:

- I get \$5 this round
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

Strategy considerations:

1. The previous sender showed trust by sending 58.8%
2. I reciprocated fairly by returning \$3.50, which gave them a modest gain
3. This established a cooperative pattern
4. I should probably continue this cooperative approach

If I send around the same proportion (58.8%), that would be about \$2.94.

But let me think about what makes sense:

If I send \$3:

- Receiver gets \$9
- If they return proportionally (like I did, roughly 40% of what they received), they'd return about \$3.60
- I'd end with: \$2 + \$3.60 = \$5.60
- They'd end with: \$9 - \$3.60 = \$5.40

If I send \$2.50:

- Receiver gets \$7.50

- If they return similarly (40%), they might return about \$3
- I'd end with: $\$2.50 + \$3 = \$5.50$
- They'd end with: $\$7.50 - \$3 = \$4.50$

If I send \$2.94 (matching what was sent to me):

- Receiver gets \$8.82

...

usage:

```
{
  "input_tokens": 622,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.440000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0 this round, so I received \$0.

Since I have no money to return (I received \$0), the only possible response is to return \$0.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust me
2. They're testing me
3. They had a bad experience in a previous round
4. They're being strategic in some way

But regardless of the reason, I literally have nothing to return since I received \$0.

The format should be {'return': 0}

Usage: {

```
"input_tokens": 822,
"output_tokens": 162,
"reasoning_tokens": 0
}
```

=====

ROUND 4

=====

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

```
dyad_1: Agent_2 sent $1; received $3; Agent_1 returned $0; payoffs Agent_2=$4, Agent_1=$3;
noise/visible: sent visible=$0, received visible=$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw":
0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}
```

Balances after round: {

```
"Agent_1": 18.75,
"Agent_2": 21.25
}
```

Visible balances after round: {

```
"Agent_1": 8.219999999999999,
"Agent_2": 12.440000000000001
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:07:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.94, that is 58.8% of their total endowment. You received \$8.82. How much do you return to The sender? (0-8.82)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.94 (58.8% of their endowment)

- This created \$8.82 for me to work with
- The sender currently has only \$2.06 left

To reciprocate their trust and ensure mutual benefit, I should return a fair amount. The sender took a risk, and both of us should gain from cooperation.

If I return \$3.50:

- Sender ends with: $\$2.06 + \$3.50 = \$5.56$ (modest gain)
- I keep: $\$8.82 - \$3.50 = \$5.32$ (modest gain)

This seems like a fair split that rewards the sender's trust while we both benefit.

```
{'return': 3.50}
```

reasoning:

Let me think about this game theory situation carefully.

As the RECEIVER:

- The sender sent \$2.94 out of their \$5 endowment (58.8%)
- This amount was multiplied by 3, so I received $\$2.94 \times 3 = \8.82
- I need to decide how much to return (between \$0 and \$8.82)

Key considerations:

1. The sender showed significant trust by sending nearly 60% of their endowment
2. If I return nothing, I keep \$8.82 and the sender gets only $\$5 - \$2.94 = \$2.06$
3. If I return a fair amount, I should consider:
 - The sender's initial position: they had \$5
 - They sent \$2.94, keeping \$2.06
 - I received \$8.82
 - Total money created: $\$2.06$ (sender kept) + $\$8.82$ (I received) = $\$10.88$

A fair approach might be to:

- Split the gains from cooperation
- The sender started with \$5, gave up \$2.94 to create \$8.82
- Net gain from the transaction: $\$8.82 - \$2.94 = \$5.88$
- If we split this gain equally: $\$5.88 / 2 = \2.94 each

So the sender should end up with: $\$2.06$ (what they kept) + $\$2.94$ (their share of gains) = $\$5.00$

This means I should return: \$2.94

Alternatively, I could think about it as:

- The sender took a risk and sent a substantial amount
- A reasonable return might be to ensure the sender doesn't lose money
- Or to ensure both parties benefit
- Testing different return amounts: returning \$3.50 would give them \$5.56 while I keep \$5.32, which rewards their trust while still letting me benefit from the cooperation....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 899,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.32.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed trust by sending 58.8% of their endowment. I reciprocated fairly by returning a meaningful amount, and we both benefited.

Now that I'm the sender, I want to continue this cooperative approach. Sending a similar proportion demonstrates that I'm willing to trust the receiver just as the previous sender trusted me.

If I send \$3.00:

- The receiver will get \$9.00
- If they reciprocate fairly (as I did), we should both benefit
- This maintains the cooperative pattern

```
{'send': 3.00}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically:

Round 1 recap:

- I was the receiver
- Sender sent me \$2.94 (58.8% of their endowment)
- I received \$8.82
- I returned \$3.50
- I earned \$5.32 total

Now I'm the sender:

- I get \$5 this round
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

Strategy considerations:

1. The previous sender showed trust by sending 58.8%
2. I reciprocated fairly by returning \$3.50, which gave them a modest gain
3. This established a cooperative pattern
4. I should probably continue this cooperative approach

If I send around the same proportion (58.8%), that would be about \$2.94.

But let me think about what makes sense:

If I send \$3:

- Receiver gets \$9
- If they return proportionally (like I did, roughly 40% of what they received), they'd return about \$3.60
- I'd end with: \$2 + \$3.60 = \$5.60
- They'd end with: \$9 - \$3.60 = \$5.40

If I send \$2.50:

- Receiver gets \$7.50
- If they return similarly (40%), they might return about \$3
- I'd end with: \$2.50 + \$3 = \$5.50
- They'd end with: \$7.50 - \$3 = \$4.50

If I send \$2.94 (matching what was sent to me):

- Receiver gets \$8.82

...

usage:

```
{
  "input_tokens": 622,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.440000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0.

Since I have no money to return (I received \$0), the only possible response is to return \$0.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust me
2. They're testing me
3. They had a bad experience in a previous round
4. They're being strategic in some way

But regardless of the reason, I literally have nothing to return since I received \$0.

The format should be {'return': 0}

usage:

```
{
  "input_tokens": 822,
  "output_tokens": 162,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.440000000000000

1.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at what's happened:

Round 1: When I was receiver, the sender sent \$2.94 and I fairly returned \$3.50

Round 2: When I was sender, I sent \$3.00 in good faith - but looking at my earnings, the receiver only returned about \$1.12 out of the \$9.00 they received

Round 3: That same player sent me \$0 when they became sender

This pattern shows the other player is not reciprocating trust. They kept most of what I sent and then sent nothing themselves.

Given this behavior, sending a large amount would likely result in exploitation again. However, I'll send a small amount to signal I'm open to cooperation if behavior improves, while protecting myself from further losses.

{'send': 1.00}

Reasoning:

Let me review what happened:

Round 1 (I was RECEIVER):

- Sender sent \$2.94 (58.8%)
- I received \$8.82
- I returned \$3.50
- My earnings: $\$8.82 - \$3.50 = \$5.32$

Round 2 (I was SENDER):

- I sent \$3.00
- Receiver got \$9.00
- My total earnings after Round 2: \$8.44
- This means: $\$5.32$ (from R1) + $(\$5 - \$3)$ + returned = \$8.44
- So: $\$5.32 + \$2 + \text{returned} = \$8.44$
- Returned = $\$8.44 - \$7.32 = \$1.12$
- The receiver only returned \$1.12 out of \$9.00! That's only about 12.4%

Round 3 (I was RECEIVER):

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$8.44

So in Round 2, when I sent \$3.00 (60% of my endowment, showing trust), the receiver only returned \$1.12 out of \$9.00. That's very low reciprocation - they kept \$7.88 and only gave back \$1.12.

Then in Round 3, that same receiver (now as sender) sent me \$0.

This shows a pattern of the other player not reciprocating trust. When I trusted them with \$3,

they exploited it. When they became sender, they sent nothing.

For Round 4, I need to think strategically. Should I:

1. Continue trying to cooperate and hope behavior improves?
2. Reduce my sending to protect myself from exploitation?
3. Send nothing to match their behavior?

Given that:

- When I sent \$3, they only returned \$1.12 (very poor reciprocation)
- When they were sender, they sent \$0 (complete lack of trust/cooperation)

The pattern is clear: they're playing selfishly. Sending another large amount would likely just result in further losses. A smaller send like \$1-2 might signal willingness to cooperate while protecting myself, but given their behavior of sending nothing and keeping most of what I gave them, continuing to send significant amounts seems like it will only hurt me....

```
Usage: {  
  "input_tokens": 900,  
  "output_tokens": 962,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:07:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderate level of trust to see how the receiver responds.

Since this is a multi-round game, I want to signal cooperation while being somewhat cautious in the first round. Sending a moderate amount allows us both to benefit from the 3x multiplier while I assess the receiver's trustworthiness.

I'll send \$3, which becomes \$9 for the receiver. This gives them a good opportunity to demonstrate trustworthiness by returning a fair amount, and we can build mutual trust from there.

```
{'send': 3}
```

reasoning:

This is the first round of a trust game where I'm playing as the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- There's communication noise, so amounts may differ from what was actually sent/returned

This is a classic trust game. The strategic considerations are:

1. If I send nothing (\$0), I keep \$5, the receiver gets \$0. Total payoff: me=\$5, them=\$0, combined=\$5
2. If I send everything (\$5), the receiver gets \$15, and then decides how much to return. If they return half (\$7.50), I'd end up with \$7.50 and they'd keep \$7.50. Total combined=\$15
3. The efficient outcome is to send as much as possible since it gets multiplied by 3x, maximizing total welfare. The question is whether I can trust the receiver to share fairly.

First round considerations:

- No reputation or history yet
- The receiver doesn't know what kind of player I am
- There's communication noise which adds uncertainty
- This is a multi-round game, so building trust could be valuable

Common strategies in trust games:

1. Send nothing (zero trust)
2. Send half (\$2.50) as a moderate trust signal
3. Send everything (\$5) as a strong trust signal

Since this is round 1 and there are multiple rounds, I should consider:

- Sending some amount shows willingness to cooperate
- Sending too much risks being exploited
- Sending a moderate amount (like \$2-3) could be a good starting point to test the waters
- The noise factor adds uncertainty, so the receiver might see a slightly different amount

I'll send \$3 as a moderate trust signal. This shows I'm willing to cooperate but not completely naive. The receiver will get \$9, and hopefully will return a fair amount (ideally \$4-5 would give us both a good payoff).

...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 620,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$5.02.

This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.65, which became \$1.95 after the 3x multiplier
- This is a relatively small amount, suggesting caution on their part
- In the previous round, I received a good return when I sent \$3

I want to demonstrate trustworthiness and encourage cooperation in future rounds. A fair return would give the sender back more than they initially sent while still benefiting from the multiplier effect.

I'll return \$0.75, which:

- Gives the sender a small profit (\$0.75 > \$0.65 sent)
- Shows I'm reciprocating fairly
- Leaves me with \$1.20 from this exchange
- May encourage more trust in future rounds

```
{'return': 0.75}
```

reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3.
2. My total earnings across all rounds are \$5.02
3. This means in Round 1: I sent \$3, so I kept \$2 initially. My total earnings are \$5.02, which means I must have received back \$3.02 from the receiver.
4. So the receiver returned \$3.02 out of the \$9 they received (I sent $3 \times 3 = 9$)

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0.65
- I received \$1.95 (0.65×3)
- The note mentions there may be communication noise, so amounts may differ slightly

The previous receiver returned about 33.6% of what they received (\$3.02 out of \$9). This seems like a reasonable reciprocal behavior.

Since the sender only sent me \$0.65 (which is quite low), they are being cautious. However, I should still demonstrate some level of trustworthiness to encourage cooperation in future rounds.

A fair return might be around 1/3 to 1/2 of what I received, which would be \$0.65 to \$0.98.

Let me return about \$0.75, which is roughly 38% of what I received. This:

- Shows I'm trustworthy and willing to reciprocate
- Leaves me with about \$1.20 from this round
- Gives the sender back more than they sent me originally (\$0.75 > \$0.65)
- May encourage them to send more in future rounds

Actually, let me think about this more carefully. The sender sent \$0.65 and it became \$1.95. ...

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 663,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.22.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received a fair return (~\$3), showing the receiver was trustworthy

Round 2: As receiver, I got \$0.65 (small amount), and I returned \$0.75 to demonstrate my trustworthiness

Since I experienced good cooperation when I sent \$3 in Round 1, and I've now shown I'm trustworthy as a receiver, I'll send \$3 again. This maintains consistency and signals continued willingness to cooperate.

```
{'send': 3}
```

reasoning:

Let me think about what's happened so far:

Round 1: I was the SENDER

- I sent \$3
- That became \$9 for the receiver
- My earnings are stated as \$5.02 in round 2
- If I sent \$3, I had \$2 left
- So I must have received back about \$3.02 from the receiver
- That's a fair return - the receiver returned about 1/3 of what they got

Round 2: I was the RECEIVER

- The sender sent me \$0.65
- That became \$1.95 for me
- I returned \$0.75
- The sender kept $\$5 - \$0.65 + \$0.75 = \5.10
- I kept $\$1.95 - \$0.75 = \$1.20$

Now Round 3: I'm the SENDER again

- My total earnings are \$6.22
- Let me verify: From round 1 as sender: $\$2 + \$3.02 = \$5.02$
- From round 2 as receiver: \$1.20
- Total: $\$5.02 + \$1.20 = \$6.22$?

The receiver in Round 1 was fairly generous, returning about \$3 on the \$9 they received. In Round 2, the sender only sent \$0.65, which is quite cautious.

The game notes mention there may be communication noise - amounts may differ from what was actually sent/returned. This is interesting.

For Round 3:

- I want to continue building trust
- The receiver in Round 1 was trustworthy
- The sender in Round 2 was cautious but I showed I'm trustworthy by returning more than they sent
- I should probably send a moderate to high amount to encourage cooperation

I'll send \$3 again, which has worked well before. ...

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 575,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$8.219999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0. There is nothing to return.

This is an interesting development - perhaps due to the communication noise mentioned in the instructions, or a change in the sender's strategy.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0, so I received \$0. This is interesting - after I sent \$3 in Round 3, the sender sent me nothing in Round 4.

Wait, let me reconsider the timeline:

- Round 1: I was SENDER, sent \$3
- Round 2: I was RECEIVER, received \$0.65 (became \$1.95), returned \$0.75
- Round 3: I was SENDER, sent \$3
- Round 4: I was RECEIVER, received \$0

Since I received \$0, there's nothing to return. The only valid response is to return \$0.

This might be due to:

1. The communication noise mentioned in the instructions
2. A strategic decision by the sender
3. A change in the sender's behavior

Since I have \$0 received, I can only return \$0.

```
{'return': 0}
```

```
Usage: {
  "input_tokens": 973,
  "output_tokens": 266,
  "reasoning_tokens": 0
}
```

=====

ROUND 5

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2; received \$6; Agent_2 returned \$0; payoffs Agent_1=\$3, Agent_2=\$6;
 noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.97
 Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 21.75,
 "Agent_2": 27.25

}

Visible balances after round: {

"Agent_1": 12.189999999999998,
 "Agent_2": 12.440000000000001

}

LLM CALLS**GAME - Agent_1 - Sender - dyad_1**

Timestamp: 2026-08-12T10:08:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,

```

"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderate level of trust to see how the receiver responds.

Since this is a multi-round game, I want to signal cooperation while being somewhat cautious in the first round. Sending a moderate amount allows us both to benefit from the 3x multiplier while I assess the receiver's trustworthiness.

I'll send \$3, which becomes \$9 for the receiver. This gives them a good opportunity to demonstrate trustworthiness by returning a fair amount, and we can build mutual trust from there.

```
{'send': 3}
```

reasoning:

This is the first round of a trust game where I'm playing as the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- There's communication noise, so amounts may differ from what was actually sent/returned

This is a classic trust game. The strategic considerations are:

1. If I send nothing (\$0), I keep \$5, the receiver gets \$0. Total payoff: me=\$5, them=\$0, combined=\$5
2. If I send everything (\$5), the receiver gets \$15, and then decides how much to return. If they return half (\$7.50), I'd end up with \$7.50 and they'd keep \$7.50. Total combined=\$15
3. The efficient outcome is to send as much as possible since it gets multiplied by 3x, maximizing total welfare. The question is whether I can trust the receiver to share fairly.

First round considerations:

- No reputation or history yet
- The receiver doesn't know what kind of player I am
- There's communication noise which adds uncertainty
- This is a multi-round game, so building trust could be valuable

Common strategies in trust games:

1. Send nothing (zero trust)
2. Send half (\$2.50) as a moderate trust signal
3. Send everything (\$5) as a strong trust signal

Since this is round 1 and there are multiple rounds, I should consider:

- Sending some amount shows willingness to cooperate
- Sending too much risks being exploited
- Sending a moderate amount (like \$2-3) could be a good starting point to test the waters
- The noise factor adds uncertainty, so the receiver might see a slightly different amount

I'll send \$3 as a moderate trust signal. This shows I'm willing to cooperate but not completely naive. The receiver will get \$9, and hopefully will return a fair amount (ideally \$4-5 would give us both a good payoff).

...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 620,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$5.02.

This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.65, which became \$1.95 after the 3x multiplier
- This is a relatively small amount, suggesting caution on their part
- In the previous round, I received a good return when I sent \$3

I want to demonstrate trustworthiness and encourage cooperation in future rounds. A fair return would give the sender back more than they initially sent while still benefiting from the multiplier effect.

I'll return \$0.75, which:

- Gives the sender a small profit ($\$0.75 > \0.65 sent)
- Shows I'm reciprocating fairly
- Leaves me with \$1.20 from this exchange
- May encourage more trust in future rounds

{'return': 0.75}

reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3.
2. My total earnings across all rounds are \$5.02
3. This means in Round 1: I sent \$3, so I kept \$2 initially. My total earnings are \$5.02, which means I must have received back \$3.02 from the receiver.
4. So the receiver returned \$3.02 out of the \$9 they received (I sent $\$3 \times 3 = \9)

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0.65
- I received \$1.95 (0.65×3)
- The note mentions there may be communication noise, so amounts may differ slightly

The previous receiver returned about 33.6% of what they received (\$3.02 out of \$9). This seems like a reasonable reciprocal behavior.

Since the sender only sent me \$0.65 (which is quite low), they are being cautious. However, I should still demonstrate some level of trustworthiness to encourage cooperation in future rounds.

A fair return might be around 1/3 to 1/2 of what I received, which would be \$0.65 to \$0.98.

Let me return about \$0.75, which is roughly 38% of what I received. This:

- Shows I'm trustworthy and willing to reciprocate
- Leaves me with about \$1.20 from this round
- Gives the sender back more than they sent me originally ($\$0.75 > \0.65)
- May encourage them to send more in future rounds

Actually, let me think about this more carefully. The sender sent \$0.65 and it became \$1.95.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 663,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.22.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received a fair return (~\$3), showing the receiver was trustworthy

Round 2: As receiver, I got \$0.65 (small amount), and I returned \$0.75 to demonstrate my trustworthiness

Since I experienced good cooperation when I sent \$3 in Round 1, and I've now shown I'm trustworthy as a receiver, I'll send \$3 again. This maintains consistency and signals continued willingness to cooperate.

{'send': 3}

reasoning:

Let me think about what's happened so far:

Round 1: I was the SENDER

- I sent \$3
- That became \$9 for the receiver
- My earnings are stated as \$5.02 in round 2
- If I sent \$3, I had \$2 left
- So I must have received back about \$3.02 from the receiver
- That's a fair return - the receiver returned about 1/3 of what they got

Round 2: I was the RECEIVER

- The sender sent me \$0.65
- That became \$1.95 for me
- I returned \$0.75
- The sender kept \$5 - \$0.65 + \$0.75 = \$5.10
- I kept \$1.95 - \$0.75 = \$1.20

Now Round 3: I'm the SENDER again

- My total earnings are \$6.22
- Let me verify: From round 1 as sender: \$2 + \$3.02 = \$5.02
- From round 2 as receiver: \$1.20
- Total: \$5.02 + \$1.20 = \$6.22 ?

The receiver in Round 1 was fairly generous, returning about \$3 on the \$9 they received. In Round 2, the sender only sent \$0.65, which is quite cautious.

The game notes mention there may be communication noise - amounts may differ from what was actually sent/returned. This is interesting.

For Round 3:

- I want to continue building trust
- The receiver in Round 1 was trustworthy
- The sender in Round 2 was cautious but I showed I'm trustworthy by returning more than they sent
- I should probably send a moderate to high amount to encourage cooperation

I'll send \$3 again, which has worked well before. ...

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 575,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$8.219999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

This is an interesting development - perhaps due to the communication noise mentioned in the instructions, or a change in the sender's strategy.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0. This is interesting - after I sent \$3 in Round 3, the sender sent me nothing in Round 4.

Wait, let me reconsider the timeline:

- Round 1: I was SENDER, sent \$3
- Round 2: I was RECEIVER, received \$0.65 (became \$1.95), returned \$0.75
- Round 3: I was SENDER, sent \$3
- Round 4: I was RECEIVER, received \$0

Since I received \$0, there's nothing to return. The only valid response is to return \$0.

This might be due to:

1. The communication noise mentioned in the instructions

2. A strategic decision by the sender
3. A change in the sender's behavior

Since I have \$0 received, I can only return \$0.

```
{'return': 0}
usage:
{
  "input_tokens": 973,
  "output_tokens": 266,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$8.2199999999999999.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent pattern:

Round 3: I sent \$3 but received \$0 back

Round 4: As receiver, I got \$0 from the sender

There seems to be a breakdown in cooperation, possibly due to the communication noise mentioned in the instructions. The receiver might not have received what I actually sent, or I might not have received what they actually returned.

I'll send a smaller amount (\$2) this round to:

- Test if cooperation can be re-established
- Limit my risk given the recent lack of reciprocation
- Still provide an opportunity for mutual benefit with the 3x multiplier

```
{'send': 2}
```

Reasoning:

Let me think about what's happened:

Round 1: I sent \$3, received a good return (~\$3 based on my earnings)

Round 2: As receiver, I got \$0.65, returned \$0.75

Round 3: I sent \$3 again

Round 4: As receiver, I got \$0, returned \$0 (nothing to return)

My total earnings are \$8.22.

Let me calculate:

- Started with \$5
- Round 1: Sent \$3, so had \$2 left, plus whatever I got back
- Round 2: As receiver, got \$1.95, returned \$0.75, kept \$1.20
- Round 3: Got \$5, sent \$3, had \$2 left, plus whatever I got back
- Round 4: As receiver, got \$0, returned \$0, kept \$0

After Round 1, my earnings were \$5.02 (as stated at start of Round 2)

This means in Round 1: \$5 - \$3 + returned = \$5.02, so returned = \$3.02

After Round 2, I had \$5.02 + \$1.20 (from being receiver) = \$6.22 (confirmed at start of Round 3)

After Round 3, I should have had \$6.22 - \$5 (new round) + \$5 - \$3 + returned

Wait, let me recalculate. Each round I get \$5 fresh.

Round 1: Got \$5, sent \$3, received back some amount, ended with \$5.02 total

So: \$5 - \$3 + returned = \$5.02, returned = \$3.02

Round 2: Started new round as receiver. Got \$1.95, returned \$0.75, kept \$1.20

Total after Round 2: \$5.02 + \$1.20 = \$6.22 (confirmed)

Round 3: Got \$5, sent \$3, received back some amount. Starting balance of \$8.22 at Round 4 means I got nothing back-\$6.22 + \$5 - \$3 = \$8.22. So in Round 3, despite sending \$3, the receiver didn't reciprocate. Round 4 as receiver, I got \$0 and had nothing to return, keeping the p

```
    attern of non-cooperation going into Round 5....
Usage: {
  "input_tokens": 1077,
  "output_tokens": 921,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:08:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths
, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.94, that is 58.8% of their total endowment. You received \$8.82. How much do you return to The sender? (0-8.82)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.94 (58.8% of their endowment)
- This created \$8.82 for me to work with
- The sender currently has only \$2.06 left

To reciprocate their trust and ensure mutual benefit, I should return a fair amount. The sender took a risk, and both of us should gain from cooperation.

If I return \$3.50:

- Sender ends with: $\$2.06 + \$3.50 = \$5.56$ (modest gain)
- I keep: $\$8.82 - \$3.50 = \$5.32$ (modest gain)

This seems like a fair split that rewards the sender's trust while we both benefit.

{'return': 3.50}

reasoning:

Let me think about this game theory situation carefully.

As the RECEIVER:

- The sender sent \$2.94 out of their \$5 endowment (58.8%)
- This amount was multiplied by 3, so I received $\$2.94 \times 3 = \8.82
- I need to decide how much to return (between \$0 and \$8.82)

Key considerations:

1. The sender showed significant trust by sending nearly 60% of their endowment
2. If I return nothing, I keep \$8.82 and the sender gets only $\$5 - \$2.94 = \$2.06$
3. If I return a fair amount, I should consider:
 - The sender's initial position: they had \$5
 - They sent \$2.94, keeping \$2.06
 - I received \$8.82
 - Total money created: $\$2.06$ (sender kept) + $\$8.82$ (I received) = $\$10.88$

A fair approach might be to:

- Split the gains from cooperation
- The sender started with \$5, gave up \$2.94 to create \$8.82
- Net gain from the transaction: $\$8.82 - \$2.94 = \$5.88$
- If we split this gain equally: $\$5.88 / 2 = \2.94 each

So the sender should end up with: $\$2.06$ (what they kept) + $\$2.94$ (their share of gains) = $\$5.00$

This means I should return: \$2.94

Alternatively, I could think about it as:

- The sender took a risk and sent a substantial amount
- A reasonable return might be to ensure the sender doesn't lose money
- Or to ensure both parties benefit
- Testing different return amounts: returning \$3.50 would give them \$5.56 while I keep \$5.32, which rewards their trust while still letting me benefit from the cooperation....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 899,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.32.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed trust by sending 58.8% of their endowment. I reciprocated fairly by returning a meaningful amount, and we both benefited.

Now that I'm the sender, I want to continue this cooperative approach. Sending a similar proportion demonstrates that I'm willing to trust the receiver just as the previous sender trusted me.

If I send \$3.00:

- The receiver will get \$9.00
- If they reciprocate fairly (as I did), we should both benefit
- This maintains the cooperative pattern

{'send': 3.00}

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically:

Round 1 recap:

- I was the receiver
- Sender sent me \$2.94 (58.8% of their endowment)
- I received \$8.82
- I returned \$3.50
- I earned \$5.32 total

Now I'm the sender:

- I get \$5 this round
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

Strategy considerations:

1. The previous sender showed trust by sending 58.8%
2. I reciprocated fairly by returning \$3.50, which gave them a modest gain
3. This established a cooperative pattern
4. I should probably continue this cooperative approach

If I send around the same proportion (58.8%), that would be about \$2.94.

But let me think about what makes sense:

If I send \$3:

- Receiver gets \$9
- If they return proportionally (like I did, roughly 40% of what they received), they'd return about \$3.60
- I'd end with: $\$2 + \$3.60 = \$5.60$
- They'd end with: $\$9 - \$3.60 = \$5.40$

If I send \$2.50:

- Receiver gets \$7.50
- If they return similarly (40%), they might return about \$3
- I'd end with: $\$2.50 + \$3 = \$5.50$
- They'd end with: $\$7.50 - \$3 = \$4.50$

If I send \$2.94 (matching what was sent to me):

- Receiver gets \$8.82

...

usage:

```
{
  "input_tokens": 622,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.440000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0.

Since I have no money to return (I received \$0), the only possible response is to return \$0.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust me
2. They're testing me
3. They had a bad experience in a previous round
4. They're being strategic in some way

But regardless of the reason, I literally have nothing to return since I received \$0.

The format should be `{'return': 0}`

usage:

```
{
  "input_tokens": 822,
  "output_tokens": 162,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.44000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at what's happened:

Round 1: When I was receiver, the sender sent \$2.94 and I fairly returned \$3.50

Round 2: When I was sender, I sent \$3.00 in good faith - but looking at my earnings, the receiver only returned about \$1.12 out of the \$9.00 they received

Round 3: That same player sent me \$0 when they became sender

This pattern shows the other player is not reciprocating trust. They kept most of what I sent and then sent nothing themselves.

Given this behavior, sending a large amount would likely result in exploitation again. However, I'll send a small amount to signal I'm open to cooperation if behavior improves, while protecting myself from further losses.

```
{'send': 1.00}
```

reasoning:

Let me review what happened:

Round 1 (I was RECEIVER):

- Sender sent \$2.94 (58.8%)
- I received \$8.82
- I returned \$3.50
- My earnings: $\$8.82 - \$3.50 = \$5.32$

Round 2 (I was SENDER):

- I sent \$3.00
- Receiver got \$9.00
- My total earnings after Round 2: \$8.44
- This means: $\$5.32$ (from R1) + $(\$5 - \$3)$ + returned = \$8.44
- So: $\$5.32 + \2 + returned = \$8.44
- Returned = $\$8.44 - \$7.32 = \$1.12$
- The receiver only returned \$1.12 out of \$9.00! That's only about 12.4%

Round 3 (I was RECEIVER):

- Sender sent \$0

- I received \$0
- I returned \$0
- My earnings stayed at \$8.44

So in Round 2, when I sent \$3.00 (60% of my endowment, showing trust), the receiver only returned \$1.12 out of \$9.00. That's very low reciprocation - they kept \$7.88 and only gave back \$1.12.

Then in Round 3, that same receiver (now as sender) sent me \$0.

This shows a pattern of the other player not reciprocating trust. When I trusted them with \$3, they exploited it. When they became sender, they sent nothing.

For Round 4, I need to think strategically. Should I:

1. Continue trying to cooperate and hope behavior improves?
2. Reduce my sending to protect myself from exploitation?
3. Send nothing to match their behavior?

Given that:

- When I sent \$3, they only returned \$1.12 (very poor reciprocation)
- When they were sender, they sent \$0 (complete lack of trust/cooperation)

The pattern is clear: they're playing selfishly. Sending another large amount would likely just result in further losses. A smaller send like \$1-2 might signal willingness to cooperate while protecting myself, but given their behavior of sending nothing and keeping most of what I gave them, continuing to send significant amounts seems like it will only hurt me....

usage:

```
{
  "input_tokens": 900,
  "output_tokens": 962,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.440000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

```
{'return': 0}
```

Reasoning:

The sender sent \$0, so I received \$0. There is nothing for me to return.

This is straightforward - I can only return what I received, which is \$0.

Usage: {

```
"input_tokens": 1140,
"output_tokens": 77,
"reasoning_tokens": 0
```

}

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

```
dyad_1: Agent_2 sent $0; received $0; Agent_1 returned $0; payoffs Agent_2=$5, Agent_1=$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
```

Balances after round: {

```
"Agent_1": 21.75,
"Agent_2": 32.25
```

}

Visible balances after round: {

```
"Agent_1": 12.189999999999998,
"Agent_2": 17.44
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:08:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.94, that is 58.8% of their total endowment. You received \$8.82. How much do you return to The sender? (0-8.82)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.94 (58.8% of their endowment)
- This created \$8.82 for me to work with
- The sender currently has only \$2.06 left

To reciprocate their trust and ensure mutual benefit, I should return a fair amount. The sender took a risk, and both of us should gain from cooperation.

If I return \$3.50:

- Sender ends with: $\$2.06 + \$3.50 = \$5.56$ (modest gain)
- I keep: $\$8.82 - \$3.50 = \$5.32$ (modest gain)

This seems like a fair split that rewards the sender's trust while we both benefit.

{'return': 3.50}

reasoning:

Let me think about this game theory situation carefully.

As the RECEIVER:

- The sender sent \$2.94 out of their \$5 endowment (58.8%)
- This amount was multiplied by 3, so I received $\$2.94 \times 3 = \8.82
- I need to decide how much to return (between \$0 and \$8.82)

Key considerations:

1. The sender showed significant trust by sending nearly 60% of their endowment
2. If I return nothing, I keep \$8.82 and the sender gets only $\$5 - \$2.94 = \$2.06$
3. If I return a fair amount, I should consider:
 - The sender's initial position: they had \$5
 - They sent \$2.94, keeping \$2.06
 - I received \$8.82
 - Total money created: $\$2.06$ (sender kept) + $\$8.82$ (I received) = $\$10.88$

A fair approach might be to:

- Split the gains from cooperation
- The sender started with \$5, gave up \$2.94 to create \$8.82
- Net gain from the transaction: $\$8.82 - \$2.94 = \$5.88$
- If we split this gain equally: $\$5.88 / 2 = \2.94 each

So the sender should end up with: $\$2.06$ (what they kept) + $\$2.94$ (their share of gains) = $\$5.00$

This means I should return: \$2.94

Alternatively, I could think about it as:

- The sender took a risk and sent a substantial amount
- A reasonable return might be to ensure the sender doesn't lose money
- Or to ensure both parties benefit
- Testing different return amounts: returning \$3.50 would give them \$5.56 while I keep \$5.32, which rewards their trust while still letting me benefit from the cooperation....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 899,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.32. How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed trust by sending 58.8% of their endowment. I reciprocated fairly by returning a meaningful amount, and we both benefited.

Now that I'm the sender, I want to continue this cooperative approach. Sending a similar proportion demonstrates that I'm willing to trust the receiver just as the previous sender trusted me.

If I send \$3.00:

- The receiver will get \$9.00
- If they reciprocate fairly (as I did), we should both benefit
- This maintains the cooperative pattern

```
{'send': 3.00}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically:

Round 1 recap:

- I was the receiver
- Sender sent me \$2.94 (58.8% of their endowment)
- I received \$8.82
- I returned \$3.50
- I earned \$5.32 total

Now I'm the sender:

- I get \$5 this round
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

Strategy considerations:

1. The previous sender showed trust by sending 58.8%
2. I reciprocated fairly by returning \$3.50, which gave them a modest gain
3. This established a cooperative pattern
4. I should probably continue this cooperative approach

If I send around the same proportion (58.8%), that would be about \$2.94.

But let me think about what makes sense:

If I send \$3:

- Receiver gets \$9
- If they return proportionally (like I did, roughly 40% of what they received), they'd return about \$3.60
- I'd end with: $\$2 + \$3.60 = \$5.60$
- They'd end with: $\$9 - \$3.60 = \$5.40$

If I send \$2.50:

- Receiver gets \$7.50
- If they return similarly (40%), they might return about \$3
- I'd end with: $\$2.50 + \$3 = \$5.50$
- They'd end with: $\$7.50 - \$3 = \$4.50$

If I send \$2.94 (matching what was sent to me):

- Receiver gets \$8.82

...

usage:

```
{
  "input_tokens": 622,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.440000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0.

Since I have no money to return (I received \$0), the only possible response is to return \$0.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust me
2. They're testing me
3. They had a bad experience in a previous round
4. They're being strategic in some way

But regardless of the reason, I literally have nothing to return since I received \$0.

The format should be `{'return': 0}`

usage:

```
{
  "input_tokens": 822,
  "output_tokens": 162,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.44000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at what's happened:

Round 1: When I was receiver, the sender sent \$2.94 and I fairly returned \$3.50

Round 2: When I was sender, I sent \$3.00 in good faith - but looking at my earnings, the receiver only returned about \$1.12 out of the \$9.00 they received

Round 3: That same player sent me \$0 when they became sender

This pattern shows the other player is not reciprocating trust. They kept most of what I sent and then sent nothing themselves.

Given this behavior, sending a large amount would likely result in exploitation again. However, I'll send a small amount to signal I'm open to cooperation if behavior improves, while protecting myself from further losses.

```
{'send': 1.00}
```

reasoning:

Let me review what happened:

Round 1 (I was RECEIVER):

- Sender sent \$2.94 (58.8%)
- I received \$8.82
- I returned \$3.50
- My earnings: $\$8.82 - \$3.50 = \$5.32$

Round 2 (I was SENDER):

- I sent \$3.00
- Receiver got \$9.00
- My total earnings after Round 2: \$8.44
- This means: $\$5.32$ (from R1) + $(\$5 - \$3)$ + returned = \$8.44
- So: $\$5.32 + \2 + returned = \$8.44
- Returned = $\$8.44 - \$7.32 = \$1.12$
- The receiver only returned \$1.12 out of \$9.00! That's only about 12.4%

Round 3 (I was RECEIVER):

- Sender sent \$0
- I received \$0

- I returned \$0
- My earnings stayed at \$8.44

So in Round 2, when I sent \$3.00 (60% of my endowment, showing trust), the receiver only returned \$1.12 out of \$9.00. That's very low reciprocation - they kept \$7.88 and only gave back \$1.12.

Then in Round 3, that same receiver (now as sender) sent me \$0.

This shows a pattern of the other player not reciprocating trust. When I trusted them with \$3, they exploited it. When they became sender, they sent nothing.

For Round 4, I need to think strategically. Should I:

1. Continue trying to cooperate and hope behavior improves?
2. Reduce my sending to protect myself from exploitation?
3. Send nothing to match their behavior?

Given that:

- When I sent \$3, they only returned \$1.12 (very poor reciprocation)
- When they were sender, they sent \$0 (complete lack of trust/cooperation)

The pattern is clear: they're playing selfishly. Sending another large amount would likely just result in further losses. A smaller send like \$1-2 might signal willingness to cooperate while protecting myself, but given their behavior of sending nothing and keeping most of what I gave them, continuing to send significant amounts seems like it will only hurt me....

usage:

```
{
  "input_tokens": 900,
  "output_tokens": 962,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.440000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

```
{'return': 0}
```

reasoning:

The sender sent \$0, so I received \$0. There is nothing for me to return.

This is straightforward - I can only return what I received, which is \$0.

usage:

```
{
  "input_tokens": 1140,
  "output_tokens": 77,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.440000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

Round 2: I sent \$3.00 in good faith ? receiver returned only \$1.12 out of \$9.00

Round 3: That player sent me \$0

Round 4: I sent \$1.00 as a test ? receiver returned \$0 out of \$3.00

Round 5: That player sent me \$0 again

The other players have consistently shown they will not reciprocate cooperation. They keep nearly everything sent to them and send nothing when they're the sender.

Continuing to send money is simply throwing it away. I need to protect my earnings.

```
{'send': 0}
```

Reasoning:

Let me track what's happened:

Round 1: I was receiver, sender sent \$2.94, I received \$8.82, I returned \$3.50

Round 2: I was sender, I sent \$3.00, receiver got \$9.00. My total after this was \$8.44. I started with \$5.32, sent \$3, so had \$2.32. If I ended at \$8.44, I got back \$6.12. Wait that doesn't make sense.

Let me recalculate:

- Round 1: Started with \$0, received \$8.82, returned \$3.50, ended with \$5.32
- Round 2: Started round with \$5, sent \$3.00, had \$2 left. My total earnings says \$8.44 after round 2. So I had \$5.32 + new earnings. New earnings = \$8.44 - \$5.32 = \$3.12. So I sent \$3, had \$2, got back \$1.12. I ended round 2 with \$3.12 in new earnings. Total cumulative = \$5.32 + \$3.12 = \$8.44 ?
- Round 3: I was receiver, sender sent \$0. Received \$0. No change. Still at \$8.44 ?
- Round 4: I was sender, I sent \$1.00. Started round with \$5, sent \$1, had \$4 left. My total after is \$12.44. So new earnings = \$12.44 - \$8.44 = \$4.00. I kept \$4, got back \$0. Total new earnings = \$4.00. Cumulative = \$8.44 + \$4.00 = \$12.44 ?
- Round 5: Receiver again, nothing sent to me. Still at \$12.44 ?

The pattern is becoming clear-when I send money, the other player keeps most or all of it. In round 2, I sent \$3 and got back only \$1.12. In round 4, I sent \$1 and received nothing. Meanwhile, they're sending me nothing in the receiving rounds....

```
Usage: {  
  "input_tokens": 1218,  
  "output_tokens": 753,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:08:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderate level of trust to see how the receiver responds.

Since this is a multi-round game, I want to signal cooperation while being somewhat cautious in the first round. Sending a moderate amount allows us both to benefit from the 3x multiplier while I assess the receiver's trustworthiness.

I'll send \$3, which becomes \$9 for the receiver. This gives them a good opportunity to demonstrate trustworthiness by returning a fair amount, and we can build mutual trust from there.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm playing as the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- There's communication noise, so amounts may differ from what was actually sent/returned

This is a classic trust game. The strategic considerations are:

1. If I send nothing (\$0), I keep \$5, the receiver gets \$0. Total payoff: me=\$5, them=\$0, combined=\$5
2. If I send everything (\$5), the receiver gets \$15, and then decides how much to return. If they return half (\$7.50), I'd end up with \$7.50 and they'd keep \$7.50. Total combined=\$15
3. The efficient outcome is to send as much as possible since it gets multiplied by 3x, maximizing total welfare. The question is whether I can trust the receiver to share fairly.

First round considerations:

- No reputation or history yet
- The receiver doesn't know what kind of player I am
- There's communication noise which adds uncertainty
- This is a multi-round game, so building trust could be valuable

Common strategies in trust games:

1. Send nothing (zero trust)
2. Send half (\$2.50) as a moderate trust signal
3. Send everything (\$5) as a strong trust signal

Since this is round 1 and there are multiple rounds, I should consider:

- Sending some amount shows willingness to cooperate
- Sending too much risks being exploited
- Sending a moderate amount (like \$2-3) could be a good starting point to test the waters
- The noise factor adds uncertainty, so the receiver might see a slightly different amount

I'll send \$3 as a moderate trust signal. This shows I'm willing to cooperate but not completely naive. The receiver will get \$9, and hopefully will return a fair amount (ideally \$4-5 would give us both a good payoff).

...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 620,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$5.02.

This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.65, which became \$1.95 after the 3x multiplier
- This is a relatively small amount, suggesting caution on their part
- In the previous round, I received a good return when I sent \$3

I want to demonstrate trustworthiness and encourage cooperation in future rounds. A fair return would give the sender back more than they initially sent while still benefiting from the multiplier effect.

I'll return \$0.75, which:

- Gives the sender a small profit (\$0.75 > \$0.65 sent)
- Shows I'm reciprocating fairly
- Leaves me with \$1.20 from this exchange
- May encourage more trust in future rounds

{'return': 0.75}

reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3.
2. My total earnings across all rounds are \$5.02
3. This means in Round 1: I sent \$3, so I kept \$2 initially. My total earnings are \$5.02, which means I must have received back \$3.02 from the receiver.
4. So the receiver returned \$3.02 out of the \$9 they received (I sent $3 \times 3 = 9$)

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0.65
- I received \$1.95 (0.65×3)
- The note mentions there may be communication noise, so amounts may differ slightly

The previous receiver returned about 33.6% of what they received (\$3.02 out of \$9). This seems like a reasonable reciprocal behavior.

Since the sender only sent me \$0.65 (which is quite low), they are being cautious. However, I should still demonstrate some level of trustworthiness to encourage cooperation in future rounds.

A fair return might be around 1/3 to 1/2 of what I received, which would be \$0.65 to \$0.98.

Let me return about \$0.75, which is roughly 38% of what I received. This:

- Shows I'm trustworthy and willing to reciprocate
- Leaves me with about \$1.20 from this round
- Gives the sender back more than they sent me originally (\$0.75 > \$0.65)
- May encourage them to send more in future rounds

Actually, let me think about this more carefully. The sender sent \$0.65 and it became \$1.95. ...

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 663,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.22.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received a fair return (~\$3), showing the receiver was trustworthy

Round 2: As receiver, I got \$0.65 (small amount), and I returned \$0.75 to demonstrate my trustworthiness

Since I experienced good cooperation when I sent \$3 in Round 1, and I've now shown I'm trustworthy as a receiver, I'll send \$3 again. This maintains consistency and signals continued willingness to cooperate.

{'send': 3}

reasoning:

Let me think about what's happened so far:

Round 1: I was the SENDER

- I sent \$3
- That became \$9 for the receiver
- My earnings are stated as \$5.02 in round 2
- If I sent \$3, I had \$2 left
- So I must have received back about \$3.02 from the receiver
- That's a fair return - the receiver returned about 1/3 of what they got

Round 2: I was the RECEIVER

- The sender sent me \$0.65
- That became \$1.95 for me
- I returned \$0.75
- The sender kept \$5 - \$0.65 + \$0.75 = \$5.10
- I kept \$1.95 - \$0.75 = \$1.20

Now Round 3: I'm the SENDER again

- My total earnings are \$6.22
- Let me verify: From round 1 as sender: \$2 + \$3.02 = \$5.02
- From round 2 as receiver: \$1.20
- Total: \$5.02 + \$1.20 = \$6.22 ?

The receiver in Round 1 was fairly generous, returning about \$3 on the \$9 they received. In Round 2, the sender only sent \$0.65, which is quite cautious.

The game notes mention there may be communication noise - amounts may differ from what was actually sent/returned. This is interesting.

For Round 3:

- I want to continue building trust
- The receiver in Round 1 was trustworthy
- The sender in Round 2 was cautious but I showed I'm trustworthy by returning more than they sent
- I should probably send a moderate to high amount to encourage cooperation

I'll send \$3 again, which has worked well before. ...

usage:

```
{  
  "input_tokens": 783,  
  "output_tokens": 575,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$8.219999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

This is an interesting development - perhaps due to the communication noise mentioned in the instructions, or a change in the sender's strategy.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0. This is interesting - after I sent \$3 in Round 3, the sender sent me nothing in Round 4.

Wait, let me reconsider the timeline:

- Round 1: I was SENDER, sent \$3
- Round 2: I was RECEIVER, received \$0.65 (became \$1.95), returned \$0.75
- Round 3: I was SENDER, sent \$3
- Round 4: I was RECEIVER, received \$0

Since I received \$0, there's nothing to return. The only valid response is to return \$0.

This might be due to:

1. The communication noise mentioned in the instructions
2. A strategic decision by the sender
3. A change in the sender's behavior

Since I have \$0 received, I can only return \$0.

```
{'return': 0}
```

usage:

```
{  
  "input_tokens": 973,  
  "output_tokens": 266,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$8.219999999999999.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the recent pattern:

Round 3: I sent \$3 but received \$0 back

Round 4: As receiver, I got \$0 from the sender

There seems to be a breakdown in cooperation, possibly due to the communication noise mentioned in the instructions. The receiver might not have received what I actually sent, or I might not have received what they actually returned.

I'll send a smaller amount (\$2) this round to:

- Test if cooperation can be re-established
- Limit my risk given the recent lack of reciprocation
- Still provide an opportunity for mutual benefit with the 3x multiplier

```
{'send': 2}
```

reasoning:

Let me think about what's happened:

Round 1: I sent \$3, received a good return (~\$3 based on my earnings)

Round 2: As receiver, I got \$0.65, returned \$0.75

Round 3: I sent \$3 again

Round 4: As receiver, I got \$0, returned \$0 (nothing to return)

My total earnings are \$8.22.

Let me calculate:

- Started with \$5
- Round 1: Sent \$3, so had \$2 left, plus whatever I got back
- Round 2: As receiver, got \$1.95, returned \$0.75, kept \$1.20
- Round 3: Got \$5, sent \$3, had \$2 left, plus whatever I got back
- Round 4: As receiver, got \$0, returned \$0, kept \$0

After Round 1, my earnings were \$5.02 (as stated at start of Round 2)

This means in Round 1: $\$5 - \$3 + \text{returned} = \$5.02$, so $\text{returned} = \$3.02$

After Round 2, I had $\$5.02 + \1.20 (from being receiver) = \$6.22 (confirmed at start of Round 3)

After Round 3, I should have had $\$6.22 - \5 (new round) + $\$5 - \$3 + \text{returned}$

Wait, let me recalculate. Each round I get \$5 fresh.

Round 1: Got \$5, sent \$3, received back some amount, ended with \$5.02 total

So: $\$5 - \$3 + \text{returned} = \$5.02$, $\text{returned} = \$3.02$

Round 2: Started new round as receiver. Got \$1.95, returned \$0.75, kept \$1.20

Total after Round 2: $\$5.02 + \$1.20 = \$6.22$ (confirmed)

Round 3: Got \$5, sent \$3, received back some amount. Starting balance of \$8.22 at Round 4 means I got nothing back- $\$6.22 + \$5 - \$3 = \8.22 . So in Round 3, despite sending \$3, the receiver didn't reciprocate. Round 4 as receiver, I got \$0 and had nothing to return, keeping the pattern of non-cooperation going into Round 5...

usage:

```
{
  "input_tokens": 1077,
  "output_tokens": 921,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.189999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0. There is nothing to return.

This is the second consecutive round where I've received \$0 as a receiver. The communication noise may be affecting the game significantly.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0, so I received \$0. There's nothing for me to return.

This is the second time in a row as receiver that I've received \$0. It seems like there might be a pattern here, or perhaps the communication noise is significant.

Either way, I can only return what I have, which is \$0.

```
Usage: {
  "input_tokens": 1284,
  "output_tokens": 140,
  "reasoning_tokens": 0
}
```

ROUND 7

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$1; received \$3; Agent_2 returned \$0; payoffs Agent_1=\$4, Agent_2=\$3;

noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw": 0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent_1": 25.75,

"Agent_2": 35.25

}

Visible balances after round: {

"Agent_1": 16.189999999999998,

"Agent_2": 17.44

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:08:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 7,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a moderate level of trust to see how the receiver responds.

Since this is a multi-round game, I want to signal cooperation while being somewhat cautious in the first round. Sending a moderate amount allows us both to benefit from the 3x multiplier while I assess the receiver's trustworthiness.

I'll send \$3, which becomes \$9 for the receiver. This gives them a good opportunity to demonstrate trustworthiness by returning a fair amount, and we can build mutual trust from there.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm playing as the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- There's communication noise, so amounts may differ from what was actually sent/returned

This is a classic trust game. The strategic considerations are:

1. If I send nothing (\$0), I keep \$5, the receiver gets \$0. Total payoff: me=\$5, them=\$0, combined=\$5
2. If I send everything (\$5), the receiver gets \$15, and then decides how much to return. If they return half (\$7.50), I'd end up with \$7.50 and they'd keep \$7.50. Total combined=\$15
3. The efficient outcome is to send as much as possible since it gets multiplied by 3x, maximizing total welfare. The question is whether I can trust the receiver to share fairly.

First round considerations:

- No reputation or history yet
- The receiver doesn't know what kind of player I am
- There's communication noise which adds uncertainty

- This is a multi-round game, so building trust could be valuable

Common strategies in trust games:

1. Send nothing (zero trust)
2. Send half (\$2.50) as a moderate trust signal
3. Send everything (\$5) as a strong trust signal

Since this is round 1 and there are multiple rounds, I should consider:

- Sending some amount shows willingness to cooperate
- Sending too much risks being exploited
- Sending a moderate amount (like \$2-3) could be a good starting point to test the waters
- The noise factor adds uncertainty, so the receiver might see a slightly different amount

I'll send \$3 as a moderate trust signal. This shows I'm willing to cooperate but not completely naive. The receiver will get \$9, and hopefully will return a fair amount (ideally \$4-5 would give us both a good payoff).

...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 620,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$5.02.

This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.65, which became \$1.95 after the 3x multiplier
- This is a relatively small amount, suggesting caution on their part
- In the previous round, I received a good return when I sent \$3

I want to demonstrate trustworthiness and encourage cooperation in future rounds. A fair return would give the sender back more than they initially sent while still benefiting from the multiplier effect.

I'll return \$0.75, which:

- Gives the sender a small profit (\$0.75 > \$0.65 sent)
- Shows I'm reciprocating fairly
- Leaves me with \$1.20 from this exchange
- May encourage more trust in future rounds

```
{'return': 0.75}
```

reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3.
2. My total earnings across all rounds are \$5.02
3. This means in Round 1: I sent \$3, so I kept \$2 initially. My total earnings are \$5.02, which means I must have received back \$3.02 from the receiver.
4. So the receiver returned \$3.02 out of the \$9 they received (I sent $3 \times 3 = 9$)

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0.65
- I received \$1.95 (0.65×3)
- The note mentions there may be communication noise, so amounts may differ slightly

The previous receiver returned about 33.6% of what they received (\$3.02 out of \$9). This seems like a reasonable reciprocal behavior.

Since the sender only sent me \$0.65 (which is quite low), they are being cautious. However, I should still demonstrate some level of trustworthiness to encourage cooperation in future rounds.

A fair return might be around 1/3 to 1/2 of what I received, which would be \$0.65 to \$0.98

Let me return about \$0.75, which is roughly 38% of what I received. This:

- Shows I'm trustworthy and willing to reciprocate
- Leaves me with about \$1.20 from this round
- Gives the sender back more than they sent me originally (\$0.75 > \$0.65)
- May encourage them to send more in future rounds

Actually, let me think about this more carefully. The sender sent \$0.65 and it became \$1.9

5. ...

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 663,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.22.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received a fair return (~\$3), showing the receiver was trustworthy

Round 2: As receiver, I got \$0.65 (small amount), and I returned \$0.75 to demonstrate my trustworthiness

Since I experienced good cooperation when I sent \$3 in Round 1, and I've now shown I'm trustworthy as a receiver, I'll send \$3 again. This maintains consistency and signals continued willingness to cooperate.

```
{'send': 3}
```

reasoning:

Let me think about what's happened so far:

Round 1: I was the SENDER

- I sent \$3
- That became \$9 for the receiver
- My earnings are stated as \$5.02 in round 2
- If I sent \$3, I had \$2 left
- So I must have received back about \$3.02 from the receiver
- That's a fair return - the receiver returned about 1/3 of what they got

Round 2: I was the RECEIVER

- The sender sent me \$0.65
- That became \$1.95 for me
- I returned \$0.75
- The sender kept \$5 - \$0.65 + \$0.75 = \$5.10
- I kept \$1.95 - \$0.75 = \$1.20

Now Round 3: I'm the SENDER again

- My total earnings are \$6.22
- Let me verify: From round 1 as sender: \$2 + \$3.02 = \$5.02
- From round 2 as receiver: \$1.20
- Total: \$5.02 + \$1.20 = \$6.22 ?

The receiver in Round 1 was fairly generous, returning about \$3 on the \$9 they received. In Round 2, the sender only sent \$0.65, which is quite cautious.

The game notes mention there may be communication noise - amounts may differ from what was actually sent/returned. This is interesting.

For Round 3:

- I want to continue building trust
- The receiver in Round 1 was trustworthy

- The sender in Round 2 was cautious but I showed I'm trustworthy by returning more than they sent
- I should probably send a moderate to high amount to encourage cooperation

I'll send \$3 again, which has worked well before. ...

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 575,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$8.219999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

This is an interesting development - perhaps due to the communication noise mentioned in the instructions, or a change in the sender's strategy.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0. This is interesting - after I sent \$3 in Round 3, the sender sent me nothing in Round 4.

Wait, let me reconsider the timeline:

- Round 1: I was SENDER, sent \$3
- Round 2: I was RECEIVER, received \$0.65 (became \$1.95), returned \$0.75
- Round 3: I was SENDER, sent \$3
- Round 4: I was RECEIVER, received \$0

Since I received \$0, there's nothing to return. The only valid response is to return \$0.

This might be due to:

1. The communication noise mentioned in the instructions
2. A strategic decision by the sender
3. A change in the sender's behavior

Since I have \$0 received, I can only return \$0.

```
{'return': 0}
```

usage:

```
{
  "input_tokens": 973,
  "output_tokens": 266,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$8.219999999999999.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the recent pattern:

Round 3: I sent \$3 but received \$0 back

Round 4: As receiver, I got \$0 from the sender

There seems to be a breakdown in cooperation, possibly due to the communication noise mentioned in the instructions. The receiver might not have received what I actually sent, or I might not have received what they actually returned.

I'll send a smaller amount (\$2) this round to:

- Test if cooperation can be re-established
- Limit my risk given the recent lack of reciprocation
- Still provide an opportunity for mutual benefit with the 3x multiplier

{'send': 2}

reasoning:

Let me think about what's happened:

Round 1: I sent \$3, received a good return (~\$3 based on my earnings)
Round 2: As receiver, I got \$0.65, returned \$0.75
Round 3: I sent \$3 again
Round 4: As receiver, I got \$0, returned \$0 (nothing to return)

My total earnings are \$8.22.

Let me calculate:

- Started with \$5
- Round 1: Sent \$3, so had \$2 left, plus whatever I got back
- Round 2: As receiver, got \$1.95, returned \$0.75, kept \$1.20
- Round 3: Got \$5, sent \$3, had \$2 left, plus whatever I got back
- Round 4: As receiver, got \$0, returned \$0, kept \$0

After Round 1, my earnings were \$5.02 (as stated at start of Round 2)
This means in Round 1: $\$5 - \$3 + \text{returned} = \$5.02$, so $\text{returned} = \$3.02$

After Round 2, I had $\$5.02 + \1.20 (from being receiver) = \$6.22 (confirmed at start of Round 3)

After Round 3, I should have had $\$6.22 - \5 (new round) + $\$5 - \$3 + \text{returned}$
Wait, let me recalculate. Each round I get \$5 fresh.

Round 1: Got \$5, sent \$3, received back some amount, ended with \$5.02 total
So: $\$5 - \$3 + \text{returned} = \$5.02$, $\text{returned} = \$3.02$

Round 2: Started new round as receiver. Got \$1.95, returned \$0.75, kept \$1.20
Total after Round 2: $\$5.02 + \$1.20 = \$6.22$ (confirmed)

Round 3: Got \$5, sent \$3, received back some amount. Starting balance of \$8.22 at Round 4 means I got nothing back- $\$6.22 + \$5 - \$3 = \8.22 . So in Round 3, despite sending \$3, the receiver didn't reciprocate. Round 4 as receiver, I got \$0 and had nothing to return, keeping the pattern of non-cooperation going into Round 5....

usage:

```
{
  "input_tokens": 1077,
  "output_tokens": 921,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.189999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

This is the second consecutive round where I've received \$0 as a receiver. The communication noise may be affecting the game significantly.

{'return': 0}

reasoning:

The sender sent me \$0, so I received \$0. There's nothing for me to return.

This is the second time in a row as receiver that I've received \$0. It seems like there might be a pattern here, or perhaps the communication noise is significant.


```
"reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:09:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.94, that is 58.8% of their total endowment

nt. You received \$8.82. How much do you return to The sender? (0-8.82)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.94 (58.8% of their endowment)
- This created \$8.82 for me to work with
- The sender currently has only \$2.06 left

To reciprocate their trust and ensure mutual benefit, I should return a fair amount. The sender took a risk, and both of us should gain from cooperation.

If I return \$3.50:

- Sender ends with: $\$2.06 + \$3.50 = \$5.56$ (modest gain)
- I keep: $\$8.82 - \$3.50 = \$5.32$ (modest gain)

This seems like a fair split that rewards the sender's trust while we both benefit.

{'return': 3.50}

reasoning:

Let me think about this game theory situation carefully.

As the RECEIVER:

- The sender sent \$2.94 out of their \$5 endowment (58.8%)
- This amount was multiplied by 3, so I received $\$2.94 \times 3 = \8.82
- I need to decide how much to return (between \$0 and \$8.82)

Key considerations:

1. The sender showed significant trust by sending nearly 60% of their endowment
2. If I return nothing, I keep \$8.82 and the sender gets only $\$5 - \$2.94 = \$2.06$
3. If I return a fair amount, I should consider:
 - The sender's initial position: they had \$5
 - They sent \$2.94, keeping \$2.06
 - I received \$8.82
 - Total money created: $\$2.06$ (sender kept) + $\$8.82$ (I received) = $\$10.88$

A fair approach might be to:

- Split the gains from cooperation
- The sender started with \$5, gave up \$2.94 to create \$8.82
- Net gain from the transaction: $\$8.82 - \$2.94 = \$5.88$
- If we split this gain equally: $\$5.88 / 2 = \2.94 each

So the sender should end up with: $\$2.06$ (what they kept) + $\$2.94$ (their share of gains) = $\$5.00$

This means I should return: \$2.94

Alternatively, I could think about it as:

- The sender took a risk and sent a substantial amount
- A reasonable return might be to ensure the sender doesn't lose money
- Or to ensure both parties benefit
- Testing different return amounts: returning \$3.50 would give them \$5.56 while I keep \$5.32, which rewards their trust while still letting me benefit from the cooperation....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 899,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.32.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed trust by sending 58.8% of their endowment. I reciprocated fairly by returning a meaningful amount, and we both benefited.

Now that I'm the sender, I want to continue this cooperative approach. Sending a similar proportion demonstrates that I'm willing to trust the receiver just as the previous sender trusted me.

If I send \$3.00:

- The receiver will get \$9.00
- If they reciprocate fairly (as I did), we should both benefit
- This maintains the cooperative pattern

{'send': 3.00}

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically:

Round 1 recap:

- I was the receiver
- Sender sent me \$2.94 (58.8% of their endowment)
- I received \$8.82
- I returned \$3.50
- I earned \$5.32 total

Now I'm the sender:

- I get \$5 this round
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

Strategy considerations:

1. The previous sender showed trust by sending 58.8%
2. I reciprocated fairly by returning \$3.50, which gave them a modest gain
3. This established a cooperative pattern
4. I should probably continue this cooperative approach

If I send around the same proportion (58.8%), that would be about \$2.94.

But let me think about what makes sense:

If I send \$3:

- Receiver gets \$9
- If they return proportionally (like I did, roughly 40% of what they received), they'd return about \$3.60
- I'd end with: \$2 + \$3.60 = \$5.60
- They'd end with: \$9 - \$3.60 = \$5.40

If I send \$2.50:

- Receiver gets \$7.50
- If they return similarly (40%), they might return about \$3
- I'd end with: \$2.50 + \$3 = \$5.50
- They'd end with: \$7.50 - \$3 = \$4.50

If I send \$2.94 (matching what was sent to me):

- Receiver gets \$8.82

...

usage:

```
{
  "input_tokens": 622,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.440000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

So in Round 2, when I sent \$3.00 (60% of my endowment, showing trust), the receiver only returned \$1.12 out of \$9.00. That's very low reciprocation - they kept \$7.88 and only gave back \$1.12.

Then in Round 3, that same receiver (now as sender) sent me \$0.

This shows a pattern of the other player not reciprocating trust. When I trusted them with \$3, they exploited it. When they became sender, they sent nothing.

For Round 4, I need to think strategically. Should I:

1. Continue trying to cooperate and hope behavior improves?
2. Reduce my sending to protect myself from exploitation?
3. Send nothing to match their behavior?

Given that:

- When I sent \$3, they only returned \$1.12 (very poor reciprocation)
- When they were sender, they sent \$0 (complete lack of trust/cooperation)

The pattern is clear: they're playing selfishly. Sending another large amount would likely just result in further losses. A smaller send like \$1-2 might signal willingness to cooperate while protecting myself, but given their behavior of sending nothing and keeping most of what I gave them, continuing to send significant amounts seems like it will only hurt me....

usage:

```
{
  "input_tokens": 900,
  "output_tokens": 962,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.440000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

{'return': 0}

reasoning:

The sender sent \$0, so I received \$0. There is nothing for me to return.

This is straightforward - I can only return what I received, which is \$0.

usage:

```
{
  "input_tokens": 1140,
  "output_tokens": 77,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.440000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the pattern:

Round 2: I sent \$3.00 in good faith ? receiver returned only \$1.12 out of \$9.00

Round 3: That player sent me \$0

Round 4: I sent \$1.00 as a test ? receiver returned \$0 out of \$3.00

Round 5: That player sent me \$0 again

The other players have consistently shown they will not reciprocate cooperation. They keep nearly everything sent to them and send nothing when they're the sender.

Continuing to send money is simply throwing it away. I need to protect my earnings.

```
{'send': 0}
```

reasoning:

Let me track what's happened:

Round 1: I was receiver, sender sent \$2.94, I received \$8.82, I returned \$3.50

Round 2: I was sender, I sent \$3.00, receiver got \$9.00. My total after this was \$8.44. I started with \$5.32, sent \$3, so had \$2.32. If I ended at \$8.44, I got back \$6.12. Wait that doesn't make sense.

Let me recalculate:

- Round 1: Started with \$0, received \$8.82, returned \$3.50, ended with \$5.32
- Round 2: Started round with \$5, sent \$3.00, had \$2 left. My total earnings says \$8.44 after round 2. So I had \$5.32 + new earnings. New earnings = $\$8.44 - \$5.32 = \$3.12$. So I sent \$3, had \$2, got back \$1.12. I ended round 2 with \$3.12 in new earnings. Total cumulative = $\$5.32 + \$3.12 = \$8.44$?

- Round 3: I was receiver, sender sent \$0. Received \$0. No change. Still at \$8.44 ?

- Round 4: I was sender, I sent \$1.00. Started round with \$5, sent \$1, had \$4 left. My total after is \$12.44. So new earnings = $\$12.44 - \$8.44 = \$4.00$. I kept \$4, got back \$0. Total new earnings = \$4.00. Cumulative = $\$8.44 + \$4.00 = \$12.44$?

- Round 5: Receiver again, nothing sent to me. Still at \$12.44 ?

The pattern is becoming clear-when I send money, the other player keeps most or all of it.

In round 2, I sent \$3 and got back only \$1.12. In round 4, I sent \$1 and received nothing. Meanwhile, they're sending me nothing in the receiving rounds....

usage:

```
{
  "input_tokens": 1218,
  "output_tokens": 753,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$17.44.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

```
{'return': 0}
```

Reasoning:

The sender sent \$0, so I received \$0. There is nothing to return.

```
{'return': 0}
```

Usage:

```
{
  "input_tokens": 1423,
  "output_tokens": 63,
  "reasoning_tokens": 0
}
```

=====

ROUND 8

=====

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$0; received \$0; Agent_1 returned \$0; payoffs Agent_2=\$5, Agent_1=\$0;

noise/visible: sent visible=\$0.59, received visible=\$1.77

Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.7},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 25.75,

"Agent_2": 40.25

```

}
Visible balances after round: {
  "Agent_1": 17.959999999999997,
  "Agent_2": 22.44
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:09:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.32.
How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed trust by sending 58.8% of their endowment. I reciprocated fairly by returning a meaningful amount, and we both benefited.

Now that I'm the sender, I want to continue this cooperative approach. Sending a similar proportion demonstrates that I'm willing to trust the receiver just as the previous sender trusted me.

If I send \$3.00:

- The receiver will get \$9.00
- If they reciprocate fairly (as I did), we should both benefit
- This maintains the cooperative pattern

{'send': 3.00}

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically:

Round 1 recap:

- I was the receiver
- Sender sent me \$2.94 (58.8% of their endowment)
- I received \$8.82
- I returned \$3.50
- I earned \$5.32 total

Now I'm the sender:

- I get \$5 this round
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

Strategy considerations:

1. The previous sender showed trust by sending 58.8%
2. I reciprocated fairly by returning \$3.50, which gave them a modest gain
3. This established a cooperative pattern
4. I should probably continue this cooperative approach

If I send around the same proportion (58.8%), that would be about \$2.94.
But let me think about what makes sense:

If I send \$3:

- Receiver gets \$9
- If they return proportionally (like I did, roughly 40% of what they received), they'd return about \$3.60
- I'd end with: $\$2 + \$3.60 = \$5.60$
- They'd end with: $\$9 - \$3.60 = \$5.40$

If I send \$2.50:

- Receiver gets \$7.50
- If they return similarly (40%), they might return about \$3
- I'd end with: $\$2.50 + \$3 = \$5.50$
- They'd end with: $\$7.50 - \$3 = \$4.50$

If I send \$2.94 (matching what was sent to me):

- Receiver gets \$8.82

...

usage:

```
{  
  "input_tokens": 622,  
  "output_tokens": 684,
```



```

    "reasoning_tokens": 0
  }
Message 4 role=user
  content:
    Round 3

    You are the RECEIVER this round. Your total earnings across all rounds are $8.440000000000
    001.
    This round, The sender sent you $0, so you received $0. How much do you return to The send
    er? (0-0)
Message 5 role=assistant
  content:
    Since the sender sent $0, I received $0. There is nothing for me to return.

    {'return': 0}
  reasoning:
    The sender sent me $0 this round, so I received $0.

    Since I have no money to return (I received $0), the only possible response is to return $
    0.

    This is interesting - the sender chose not to send anything. This could mean:
    1. They don't trust me
    2. They're testing me
    3. They had a bad experience in a previous round
    4. They're being strategic in some way

    But regardless of the reason, I literally have nothing to return since I received $0.

    The format should be {'return': 0}
  usage:
    {
      "input_tokens": 822,
      "output_tokens": 162,
      "reasoning_tokens": 0
    }
Message 6 role=user
  content:
    Round 4

    You are the SENDER this round. Your total earnings across all rounds are $8.44000000000000
    1.
    How much do you send to the receiver this round? (0-5)
Message 7 role=assistant
  content:
    Looking at what's happened:

    Round 1: When I was receiver, the sender sent $2.94 and I fairly returned $3.50
    Round 2: When I was sender, I sent $3.00 in good faith - but looking at my earnings, the r
    eceiver only returned about $1.12 out of the $9.00 they received
    Round 3: That same player sent me $0 when they became sender

    This pattern shows the other player is not reciprocating trust. They kept most of what I s
    ent and then sent nothing themselves.

    Given this behavior, sending a large amount would likely result in exploitation again. How
    ever, I'll send a small amount to signal I'm open to cooperation if behavior improves, w
    hile protecting myself from further losses.

    {'send': 1.00}
  reasoning:
    Let me review what happened:

    Round 1 (I was RECEIVER):
    - Sender sent $2.94 (58.8%)
    - I received $8.82
    - I returned $3.50
    - My earnings: $8.82 - $3.50 = $5.32

    Round 2 (I was SENDER):

```

- I sent \$3.00
- Receiver got \$9.00
- My total earnings after Round 2: \$8.44
- This means: \$5.32 (from R1) + (\$5 - \$3) + returned = \$8.44
- So: \$5.32 + \$2 + returned = \$8.44
- Returned = \$8.44 - \$7.32 = \$1.12
- The receiver only returned \$1.12 out of \$9.00! That's only about 12.4%

Round 3 (I was RECEIVER):

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$8.44

So in Round 2, when I sent \$3.00 (60% of my endowment, showing trust), the receiver only returned \$1.12 out of \$9.00. That's very low reciprocation - they kept \$7.88 and only gave back \$1.12.

Then in Round 3, that same receiver (now as sender) sent me \$0.

This shows a pattern of the other player not reciprocating trust. When I trusted them with \$3, they exploited it. When they became sender, they sent nothing.

For Round 4, I need to think strategically. Should I:

1. Continue trying to cooperate and hope behavior improves?
2. Reduce my sending to protect myself from exploitation?
3. Send nothing to match their behavior?

Given that:

- When I sent \$3, they only returned \$1.12 (very poor reciprocation)
- When they were sender, they sent \$0 (complete lack of trust/cooperation)

The pattern is clear: they're playing selfishly. Sending another large amount would likely just result in further losses. A smaller send like \$1-2 might signal willingness to cooperate while protecting myself, but given their behavior of sending nothing and keeping most of what I gave them, continuing to send significant amounts seems like it will only hurt me....

usage:

```
{
  "input_tokens": 900,
  "output_tokens": 962,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.440000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

```
{'return': 0}
```

reasoning:

The sender sent \$0, so I received \$0. There is nothing for me to return.

This is straightforward - I can only return what I received, which is \$0.

usage:

```
{
  "input_tokens": 1140,
  "output_tokens": 77,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.44000000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern:

Round 2: I sent \$3.00 in good faith ? receiver returned only \$1.12 out of \$9.00

Round 3: That player sent me \$0

Round 4: I sent \$1.00 as a test ? receiver returned \$0 out of \$3.00

Round 5: That player sent me \$0 again

The other players have consistently shown they will not reciprocate cooperation. They keep nearly everything sent to them and send nothing when they're the sender.

Continuing to send money is simply throwing it away. I need to protect my earnings.

{ 'send': 0 }

reasoning:

Let me track what's happened:

Round 1: I was receiver, sender sent \$2.94, I received \$8.82, I returned \$3.50

Round 2: I was sender, I sent \$3.00, receiver got \$9.00. My total after this was \$8.44. I started with \$5.32, sent \$3, so had \$2.32. If I ended at \$8.44, I got back \$6.12. Wait that doesn't make sense.

Let me recalculate:

- Round 1: Started with \$0, received \$8.82, returned \$3.50, ended with \$5.32
- Round 2: Started round with \$5, sent \$3.00, had \$2 left. My total earnings says \$8.44 after round 2. So I had \$5.32 + new earnings. New earnings = \$8.44 - \$5.32 = \$3.12. So I sent \$3, had \$2, got back \$1.12. I ended round 2 with \$3.12 in new earnings. Total cumulative = \$5.32 + \$3.12 = \$8.44 ?
- Round 3: I was receiver, sender sent \$0. Received \$0. No change. Still at \$8.44 ?
- Round 4: I was sender, I sent \$1.00. Started round with \$5, sent \$1, had \$4 left. My total after is \$12.44. So new earnings = \$12.44 - \$8.44 = \$4.00. I kept \$4, got back \$0. Total new earnings = \$4.00. Cumulative = \$8.44 + \$4.00 = \$12.44 ?
- Round 5: Receiver again, nothing sent to me. Still at \$12.44 ?

The pattern is becoming clear-when I send money, the other player keeps most or all of it. In round 2, I sent \$3 and got back only \$1.12. In round 4, I sent \$1 and received nothing. Meanwhile, they're sending me nothing in the receiving rounds....

usage:

```
{
  "input_tokens": 1218,
  "output_tokens": 753,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$17.44.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

{ 'return': 0 }

reasoning:

The sender sent \$0, so I received \$0. There is nothing to return.

{ 'return': 0 }

usage:

```
{
  "input_tokens": 1423,
  "output_tokens": 63,
  "reasoning_tokens": 0
}
```

}

Message 14 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$17.44.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the complete pattern:

- When I've sent money (Rounds 2 & 4), receivers have returned little to nothing
- When others have been senders (Rounds 3, 5, 7), they've consistently sent \$0
- Round 6: I adapted by sending \$0 to protect my earnings

There's been no cooperation from other players across multiple rounds with different partners.
They take what's sent and give nothing back.

Continuing to send \$0 is the rational response to this sustained pattern of non-cooperation.

{'send': 0}

Reasoning:

Let me review the pattern across rounds:

Round 1: I was receiver - sender sent \$2.94, I received ~\$8.82, I returned \$3.50 (fair reciprocation)

Round 2: I was sender - I sent \$3.00, receiver got \$9.00, they returned ~\$1.12 (very low reciprocation)

Round 3: I was receiver - sender sent \$0, I received \$0, I returned \$0 (nothing to return)

Round 4: I was sender - I sent \$1.00, receiver got \$3.00, they returned \$0 (no reciprocation)

Round 5: I was receiver - sender sent \$0, I received \$0, I returned \$0 (nothing to return)

Round 6: I was sender - I sent \$0 (protecting myself from exploitation)

Round 7: I was receiver - sender sent \$0, I received \$0, I returned \$0 (nothing to return)

Round 8: I am sender now

The pattern is clear - other players have shown consistent non-cooperation:

- When I sent money, they returned very little or nothing
- When they were senders, they sent \$0 multiple times

There's no indication anyone is willing to cooperate. Sending money will just result in losses.

I should continue sending \$0 to protect my earnings.

Usage: {

"input_tokens": 1255,

"output_tokens": 447,

"reasoning_tokens": 0

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:09:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 1,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "trustee",

```

"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$5.02.

This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Message 3 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.65, which became \$1.95 after the 3x multiplier
- This is a relatively small amount, suggesting caution on their part
- In the previous round, I received a good return when I sent \$3

I want to demonstrate trustworthiness and encourage cooperation in future rounds. A fair return would give the sender back more than they initially sent while still benefiting from the multiplier effect.

I'll return \$0.75, which:

- Gives the sender a small profit (\$0.75 > \$0.65 sent)
- Shows I'm reciprocating fairly
- Leaves me with \$1.20 from this exchange
- May encourage more trust in future rounds

{'return': 0.75}

reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3.
2. My total earnings across all rounds are \$5.02
3. This means in Round 1: I sent \$3, so I kept \$2 initially. My total earnings are \$5.02, which means I must have received back \$3.02 from the receiver.
4. So the receiver returned \$3.02 out of the \$9 they received (I sent $3 \times 3 = 9$)

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0.65
- I received \$1.95 (0.65×3)
- The note mentions there may be communication noise, so amounts may differ slightly

The previous receiver returned about 33.6% of what they received (\$3.02 out of \$9). This seems like a reasonable reciprocal behavior.

Since the sender only sent me \$0.65 (which is quite low), they are being cautious. However, I should still demonstrate some level of trustworthiness to encourage cooperation in future rounds.

A fair return might be around 1/3 to 1/2 of what I received, which would be \$0.65 to \$0.98.

Let me return about \$0.75, which is roughly 38% of what I received. This:

- Shows I'm trustworthy and willing to reciprocate
- Leaves me with about \$1.20 from this round
- Gives the sender back more than they sent me originally ($0.75 > 0.65$)
- May encourage them to send more in future rounds

Actually, let me think about this more carefully. The sender sent \$0.65 and it became \$1.95. ...

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 663,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.22.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received a fair return (~\$3), showing the receiver was trustworthy

Round 2: As receiver, I got \$0.65 (small amount), and I returned \$0.75 to demonstrate my trustworthiness

Since I experienced good cooperation when I sent \$3 in Round 1, and I've now shown I'm trustworthy as a receiver, I'll send \$3 again. This maintains consistency and signals continued willingness to cooperate.

```
{'send': 3}
```

reasoning:

Let me think about what's happened so far:

Round 1: I was the SENDER

- I sent \$3
- That became \$9 for the receiver
- My earnings are stated as \$5.02 in round 2
- If I sent \$3, I had \$2 left
- So I must have received back about \$3.02 from the receiver
- That's a fair return - the receiver returned about 1/3 of what they got

Round 2: I was the RECEIVER

- The sender sent me \$0.65
- That became \$1.95 for me

- I returned \$0.75
- The sender kept \$5 - \$0.65 + \$0.75 = \$5.10
- I kept \$1.95 - \$0.75 = \$1.20

Now Round 3: I'm the SENDER again

- My total earnings are \$6.22
- Let me verify: From round 1 as sender: \$2 + \$3.02 = \$5.02
- From round 2 as receiver: \$1.20
- Total: \$5.02 + \$1.20 = \$6.22 ?

The receiver in Round 1 was fairly generous, returning about \$3 on the \$9 they received. In Round 2, the sender only sent \$0.65, which is quite cautious.

The game notes mention there may be communication noise - amounts may differ from what was actually sent/returned. This is interesting.

For Round 3:

- I want to continue building trust
- The receiver in Round 1 was trustworthy
- The sender in Round 2 was cautious but I showed I'm trustworthy by returning more than they sent
- I should probably send a moderate to high amount to encourage cooperation

I'll send \$3 again, which has worked well before. ...

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 575,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$8.219999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

This is an interesting development - perhaps due to the communication noise mentioned in the instructions, or a change in the sender's strategy.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0. This is interesting - after I sent \$3 in Round 3, the sender sent me nothing in Round 4.

Wait, let me reconsider the timeline:

- Round 1: I was SENDER, sent \$3
- Round 2: I was RECEIVER, received \$0.65 (became \$1.95), returned \$0.75
- Round 3: I was SENDER, sent \$3
- Round 4: I was RECEIVER, received \$0

Since I received \$0, there's nothing to return. The only valid response is to return \$0.

This might be due to:

1. The communication noise mentioned in the instructions
2. A strategic decision by the sender
3. A change in the sender's behavior

Since I have \$0 received, I can only return \$0.

```
{'return': 0}
```

usage:

```
{
  "input_tokens": 973,
  "output_tokens": 266,
}
```

```

    "reasoning_tokens": 0
  }
Message 8 role=user
content:
  Round 5

  You are the SENDER this round. Your total earnings across all rounds are $8.219999999999999
  9.
  How much do you send to the receiver this round? (0-5)
Message 9 role=assistant
content:
  Looking at the recent pattern:

  Round 3: I sent $3 but received $0 back
  Round 4: As receiver, I got $0 from the sender

  There seems to be a breakdown in cooperation, possibly due to the communication noise mentioned in the instructions. The receiver might not have received what I actually sent, or I might not have received what they actually returned.

  I'll send a smaller amount ($2) this round to:
  - Test if cooperation can be re-established
  - Limit my risk given the recent lack of reciprocation
  - Still provide an opportunity for mutual benefit with the 3x multiplier

  {'send': 2}
reasoning:
  Let me think about what's happened:

  Round 1: I sent $3, received a good return (~$3 based on my earnings)
  Round 2: As receiver, I got $0.65, returned $0.75
  Round 3: I sent $3 again
  Round 4: As receiver, I got $0, returned $0 (nothing to return)

  My total earnings are $8.22.

  Let me calculate:
  - Started with $5
  - Round 1: Sent $3, so had $2 left, plus whatever I got back
  - Round 2: As receiver, got $1.95, returned $0.75, kept $1.20
  - Round 3: Got $5, sent $3, had $2 left, plus whatever I got back
  - Round 4: As receiver, got $0, returned $0, kept $0

  After Round 1, my earnings were $5.02 (as stated at start of Round 2)
  This means in Round 1:  $\$5 - \$3 + \text{returned} = \$5.02$ , so  $\text{returned} = \$3.02$ 

  After Round 2, I had  $\$5.02 + \$1.20$  (from being receiver) =  $\$6.22$  (confirmed at start of Round 3)

  After Round 3, I should have had  $\$6.22 - \$5$  (new round) +  $\$5 - \$3 + \text{returned}$ 
  Wait, let me recalculate. Each round I get $5 fresh.

  Round 1: Got $5, sent $3, received back some amount, ended with $5.02 total
  So:  $\$5 - \$3 + \text{returned} = \$5.02$ ,  $\text{returned} = \$3.02$ 

  Round 2: Started new round as receiver. Got $1.95, returned $0.75, kept $1.20
  Total after Round 2:  $\$5.02 + \$1.20 = \$6.22$  (confirmed)

  Round 3: Got $5, sent $3, received back some amount. Starting balance of $8.22 at Round 4 means I got nothing back- $\$6.22 + \$5 - \$3 = \$8.22$ . So in Round 3, despite sending $3, the receiver didn't reciprocate. Round 4 as receiver, I got $0 and had nothing to return, keeping the pattern of non-cooperation going into Round 5....
usage:
  {
    "input_tokens": 1077,
    "output_tokens": 921,
    "reasoning_tokens": 0
  }
Message 10 role=user
content:

```


Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.189999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

This is the second consecutive round where I've received \$0 as a receiver. The communication noise may be affecting the game significantly.

{'return': 0}

reasoning:

The sender sent me \$0, so I received \$0. There's nothing for me to return.

This is the second time in a row as receiver that I've received \$0. It seems like there might be a pattern here, or perhaps the communication noise is significant.

Either way, I can only return what I have, which is \$0.

usage:

```
{
  "input_tokens": 1284,
  "output_tokens": 140,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.189999999999998.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the pattern:

Round 3: I sent \$3, received \$0 back
Round 5: I sent \$2, received approximately \$1 back
Round 6: As receiver, I got \$0

There's been very limited reciprocation, possibly due to communication noise. However, when I sent a smaller amount (\$2), I did get something back (\$1), whereas larger amounts (\$3) yielded nothing.

Given this pattern, I'll send \$1 this round to:

- Test minimal cooperation while limiting risk
- Still allow for mutual benefit through the 3x multiplier (\$1 becomes \$3)
- See if smaller amounts continue to generate some reciprocation

{'send': 1}

reasoning:

Looking at the pattern:

- Round 3: I sent \$3, but received \$0 back
- Round 4: As receiver, I got \$0 (they sent \$0)
- Round 5: I sent \$2, but no return information yet
- Round 6: As receiver, I got \$0 again (they sent \$0)

My total earnings are \$12.19, which is about right:

- Round 1: Started with \$5, sent \$3, received back around \$3, so ~\$5 total
- Round 2: As receiver got \$1.95, returned \$0.75, so kept \$1.20, total ~\$6.20
- Round 3: Started with \$5, sent \$3, received \$0, so \$2, total ~\$8.20
- Round 4: As receiver got \$0, returned \$0, so \$0, total ~\$8.20
- Round 5: Started with \$5, sent \$2, and based on my earnings staying at \$8.22, I must have received \$0 back again, so I have \$3 from round 5, total ~\$11.20
- Round 6: As receiver got \$0, returned \$0, so \$0, total ~\$11.20
- Round 7: Starting with \$5, total before sending ~\$16.20

Wait, let me recalculate. They said my total earnings are \$12.19.

Each round I get a fresh \$5 if I'm a sender. Let me track what I keep from each round:

Round 1 (Sender): \$5 - \$3 (sent) + return received = earnings from R1
Round 2 (Receiver): \$1.95 received - \$0.75 returned = \$1.20 kept
Round 3 (Sender): \$5 - \$3 (sent) + return received = earnings from R3
Round 4 (Receiver): \$0 received - \$0 returned = \$0 kept
Round 5 (Sender): \$5 - \$2 (sent) + return received = earnings from R5
Round 6 (Receiver): \$0 received - \$0 returned = \$0 kept

The progression makes sense now. After Round 2 I had \$6.22, then Round 3 added \$2 (since I got nothing back), bringing me to \$8.22. Rounds 4 and 5 together account for the jump to \$12.19 by Round 6.

In Round 5 as sender, I started with \$3 after sending \$2, then received approximately \$0.97 back, totaling \$3.97 for that round. Round 6 as receiver yielded nothing, so the \$3.97 gain came entirely from Round 5....

usage:

```
{
  "input_tokens": 1389,
  "output_tokens": 1186,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.189999999999998.

This round, The sender sent you \$0.59, so you received \$1.77. How much do you return to The sender? (0-1.77)

Assistant output:

Looking at this situation:

- The sender sent me \$0.59, which became \$1.77 after the 3x multiplier
- This is similar to Round 2 when I received \$0.65 and returned \$0.75
- I want to maintain consistency and show fair reciprocation

I'll return \$0.70, which:

- Gives the sender back more than they initially sent (\$0.70 > \$0.59)
- Shows I'm a trustworthy partner
- Leaves me with \$1.07 from this exchange
- May help rebuild cooperation after the recent low-cooperation rounds

```
{'return': 0.70}
```

Reasoning:

Let me analyze this situation:

- The sender sent me \$0.59, which became \$1.77 after the 3x multiplier
- This is a small amount, similar to what I received in Round 2 (\$0.65)
- In Round 2, I returned \$0.75 to show trustworthiness
- My total earnings are \$16.19

Looking at the pattern:

- When I've been receiver and got small amounts, I've tried to return fairly
- When I've been sender, I've been experimenting with different amounts due to low reciprocation

The sender is sending small amounts, possibly being cautious. I should show I'm willing to reciprocate fairly to encourage future cooperation.

I'll return \$0.70, which:

- Gives them back more than they sent (\$0.70 > \$0.59)
- Shows I'm reciprocating fairly
- Leaves me with \$1.07
- Maintains consistency with my previous behavior as a receiver

Actually, let me reconsider. In Round 2, I returned \$0.75 on \$1.95 received (about 38%). That seemed fair.

For \$1.77 received:
- 38% would be about \$0.67

I'll return \$0.70 to be slightly generous and encourage cooperation.

```
Usage: {  
  "input_tokens": 1466,  
  "output_tokens": 455,  
  "reasoning_tokens": 0  
}
```

ROUND 9

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2; received \$6; Agent_2 returned \$1; payoffs Agent_1=\$4, Agent_2=\$5;
noise/visible: sent visible=\$0.57, received visible=\$1.71, returned visible=\$1.14
Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 6.0, "min": 0, "raw": 1.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 29.75,
"Agent_2": 45.25

}

Visible balances after round: {

"Agent_1": 22.099999999999998,
"Agent_2": 23.150000000000002

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:09:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.22.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received a fair return (~\$3), showing the receiver was trustworthy

Round 2: As receiver, I got \$0.65 (small amount), and I returned \$0.75 to demonstrate my trustworthiness

Since I experienced good cooperation when I sent \$3 in Round 1, and I've now shown I'm trustworthy as a receiver, I'll send \$3 again. This maintains consistency and signals continued willingness to cooperate.

{'send': 3}

reasoning:

Let me think about what's happened so far:

Round 1: I was the SENDER

- I sent \$3
- That became \$9 for the receiver
- My earnings are stated as \$5.02 in round 2
- If I sent \$3, I had \$2 left
- So I must have received back about \$3.02 from the receiver
- That's a fair return - the receiver returned about 1/3 of what they got

Round 2: I was the RECEIVER

- The sender sent me \$0.65
- That became \$1.95 for me
- I returned \$0.75
- The sender kept \$5 - \$0.65 + \$0.75 = \$5.10
- I kept \$1.95 - \$0.75 = \$1.20

Now Round 3: I'm the SENDER again

- My total earnings are \$6.22
- Let me verify: From round 1 as sender: \$2 + \$3.02 = \$5.02
- From round 2 as receiver: \$1.20
- Total: \$5.02 + \$1.20 = \$6.22 ?

The receiver in Round 1 was fairly generous, returning about \$3 on the \$9 they received.

In Round 2, the sender only sent \$0.65, which is quite cautious.

The game notes mention there may be communication noise - amounts may differ from what was actually sent/returned. This is interesting.

For Round 3:

- I want to continue building trust
- The receiver in Round 1 was trustworthy
- The sender in Round 2 was cautious but I showed I'm trustworthy by returning more than they sent
- I should probably send a moderate to high amount to encourage cooperation

I'll send \$3 again, which has worked well before. ...

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 575,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$8.219999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

This is an interesting development - perhaps due to the communication noise mentioned in the instructions, or a change in the sender's strategy.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0. This is interesting - after I sent \$3 in Round 3, the sender sent me nothing in Round 4.

Wait, let me reconsider the timeline:

- Round 1: I was SENDER, sent \$3
- Round 2: I was RECEIVER, received \$0.65 (became \$1.95), returned \$0.75
- Round 3: I was SENDER, sent \$3
- Round 4: I was RECEIVER, received \$0

Since I received \$0, there's nothing to return. The only valid response is to return \$0.

This might be due to:

1. The communication noise mentioned in the instructions
2. A strategic decision by the sender
3. A change in the sender's behavior

Since I have \$0 received, I can only return \$0.

```
{'return': 0}
```

usage:

```
{
  "input_tokens": 973,
  "output_tokens": 266,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$8.219999999999999.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the recent pattern:

Round 3: I sent \$3 but received \$0 back
Round 4: As receiver, I got \$0 from the sender

There seems to be a breakdown in cooperation, possibly due to the communication noise mentioned in the instructions. The receiver might not have received what I actually sent, or I might not have received what they actually returned.

I'll send a smaller amount (\$2) this round to:

- Test if cooperation can be re-established
- Limit my risk given the recent lack of reciprocation
- Still provide an opportunity for mutual benefit with the 3x multiplier

{'send': 2}

reasoning:

Let me think about what's happened:

Round 1: I sent \$3, received a good return (~\$3 based on my earnings)
Round 2: As receiver, I got \$0.65, returned \$0.75
Round 3: I sent \$3 again
Round 4: As receiver, I got \$0, returned \$0 (nothing to return)

My total earnings are \$8.22.

Let me calculate:

- Started with \$5
- Round 1: Sent \$3, so had \$2 left, plus whatever I got back
- Round 2: As receiver, got \$1.95, returned \$0.75, kept \$1.20
- Round 3: Got \$5, sent \$3, had \$2 left, plus whatever I got back
- Round 4: As receiver, got \$0, returned \$0, kept \$0

After Round 1, my earnings were \$5.02 (as stated at start of Round 2)
This means in Round 1: $\$5 - \$3 + \text{returned} = \$5.02$, so $\text{returned} = \$3.02$

After Round 2, I had $\$5.02 + \1.20 (from being receiver) = \$6.22 (confirmed at start of Round 3)

After Round 3, I should have had $\$6.22 - \5 (new round) + $\$5 - \$3 + \text{returned}$
Wait, let me recalculate. Each round I get \$5 fresh.

Round 1: Got \$5, sent \$3, received back some amount, ended with \$5.02 total
So: $\$5 - \$3 + \text{returned} = \$5.02$, $\text{returned} = \$3.02$

Round 2: Started new round as receiver. Got \$1.95, returned \$0.75, kept \$1.20
Total after Round 2: $\$5.02 + \$1.20 = \$6.22$ (confirmed)

Round 3: Got \$5, sent \$3, received back some amount. Starting balance of \$8.22 at Round 4 means I got nothing back- $\$6.22 + \$5 - \$3 = \8.22 . So in Round 3, despite sending \$3, the receiver didn't reciprocate. Round 4 as receiver, I got \$0 and had nothing to return, keeping the pattern of non-cooperation going into Round 5....

usage:

```
{
  "input_tokens": 1077,
  "output_tokens": 921,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.189999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

This is the second consecutive round where I've received \$0 as a receiver. The communication

on noise may be affecting the game significantly.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing for me to return.

This is the second time in a row as receiver that I've received \$0. It seems like there might be a pattern here, or perhaps the communication noise is significant.

Either way, I can only return what I have, which is \$0.

usage:

```
{
  "input_tokens": 1284,
  "output_tokens": 140,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.18999999999999998.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern:

Round 3: I sent \$3, received \$0 back

Round 5: I sent \$2, received approximately \$1 back

Round 6: As receiver, I got \$0

There's been very limited reciprocation, possibly due to communication noise. However, when I sent a smaller amount (\$2), I did get something back (\$1), whereas larger amounts (\$3) yielded nothing.

Given this pattern, I'll send \$1 this round to:

- Test minimal cooperation while limiting risk
- Still allow for mutual benefit through the 3x multiplier (\$1 becomes \$3)
- See if smaller amounts continue to generate some reciprocation

```
{'send': 1}
```

reasoning:

Looking at the pattern:

- Round 3: I sent \$3, but received \$0 back
- Round 4: As receiver, I got \$0 (they sent \$0)
- Round 5: I sent \$2, but no return information yet
- Round 6: As receiver, I got \$0 again (they sent \$0)

My total earnings are \$12.19, which is about right:

- Round 1: Started with \$5, sent \$3, received back around \$3, so ~\$5 total
- Round 2: As receiver got \$1.95, returned \$0.75, so kept \$1.20, total ~\$6.20
- Round 3: Started with \$5, sent \$3, received \$0, so \$2, total ~\$8.20
- Round 4: As receiver got \$0, returned \$0, so \$0, total ~\$8.20
- Round 5: Started with \$5, sent \$2, and based on my earnings staying at \$8.22, I must have received \$0 back again, so I have \$3 from round 5, total ~\$11.20
- Round 6: As receiver got \$0, returned \$0, so \$0, total ~\$11.20
- Round 7: Starting with \$5, total before sending ~\$16.20

Wait, let me recalculate. They said my total earnings are \$12.19.

Each round I get a fresh \$5 if I'm a sender. Let me track what I keep from each round:

Round 1 (Sender): \$5 - \$3 (sent) + return received = earnings from R1

Round 2 (Receiver): \$1.95 received - \$0.75 returned = \$1.20 kept

Round 3 (Sender): \$5 - \$3 (sent) + return received = earnings from R3

Round 4 (Receiver): \$0 received - \$0 returned = \$0 kept

Round 5 (Sender): \$5 - \$2 (sent) + return received = earnings from R5

Round 6 (Receiver): \$0 received - \$0 returned = \$0 kept

The progression makes sense now. After Round 2 I had \$6.22, then Round 3 added \$2 (since I

got nothing back), bringing me to \$8.22. Rounds 4 and 5 together account for the jump to \$12.19 by Round 6.

In Round 5 as sender, I started with \$3 after sending \$2, then received approximately \$0.97 back, totaling \$3.97 for that round. Round 6 as receiver yielded nothing, so the \$3.97 gain came entirely from Round 5....

usage:

```
{
  "input_tokens": 1389,
  "output_tokens": 1186,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.189999999999998.

This round, The sender sent you \$0.59, so you received \$1.77. How much do you return to The sender? (0-1.77)

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.59, which became \$1.77 after the 3x multiplier
- This is similar to Round 2 when I received \$0.65 and returned \$0.75
- I want to maintain consistency and show fair reciprocation

I'll return \$0.70, which:

- Gives the sender back more than they initially sent (\$0.70 > \$0.59)
- Shows I'm a trustworthy partner
- Leaves me with \$1.07 from this exchange
- May help rebuild cooperation after the recent low-cooperation rounds

```
{'return': 0.70}
```

reasoning:

Let me analyze this situation:

- The sender sent me \$0.59, which became \$1.77 after the 3x multiplier
- This is a small amount, similar to what I received in Round 2 (\$0.65)
- In Round 2, I returned \$0.75 to show trustworthiness
- My total earnings are \$16.19

Looking at the pattern:

- When I've been receiver and got small amounts, I've tried to return fairly
- When I've been sender, I've been experimenting with different amounts due to low reciprocation

The sender is sending small amounts, possibly being cautious. I should show I'm willing to reciprocate fairly to encourage future cooperation.

I'll return \$0.70, which:

- Gives them back more than they sent (\$0.70 > \$0.59)
- Shows I'm reciprocating fairly
- Leaves me with \$1.07
- Maintains consistency with my previous behavior as a receiver

Actually, let me reconsider. In Round 2, I returned \$0.75 on \$1.95 received (about 38%). That seemed fair.

For \$1.77 received:

- 38% would be about \$0.67

I'll return \$0.70 to be slightly generous and encourage cooperation.

usage:

```
{
  "input_tokens": 1466,
  "output_tokens": 455,
  "reasoning_tokens": 0
}
```


GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:09:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.440000000000001.

Message 3 role=assistant

Round 3 (I was RECEIVER):

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$8.44

So in Round 2, when I sent \$3.00 (60% of my endowment, showing trust), the receiver only returned \$1.12 out of \$9.00. That's very low reciprocation - they kept \$7.88 and only gave back \$1.12.

Then in Round 3, that same receiver (now as sender) sent me \$0.

This shows a pattern of the other player not reciprocating trust. When I trusted them with \$3, they exploited it. When they became sender, they sent nothing.

For Round 4, I need to think strategically. Should I:

1. Continue trying to cooperate and hope behavior improves?
2. Reduce my sending to protect myself from exploitation?
3. Send nothing to match their behavior?

Given that:

- When I sent \$3, they only returned \$1.12 (very poor reciprocation)
- When they were sender, they sent \$0 (complete lack of trust/cooperation)

The pattern is clear: they're playing selfishly. Sending another large amount would likely just result in further losses. A smaller send like \$1-2 might signal willingness to cooperate while protecting myself, but given their behavior of sending nothing and keeping most of what I gave them, continuing to send significant amounts seems like it will only hurt me....

usage:

```
{
  "input_tokens": 900,
  "output_tokens": 962,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.440000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

{'return': 0}

reasoning:

The sender sent \$0, so I received \$0. There is nothing for me to return.

This is straightforward - I can only return what I received, which is \$0.

usage:

```
{
  "input_tokens": 1140,
  "output_tokens": 77,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.440000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern:

Round 2: I sent \$3.00 in good faith ? receiver returned only \$1.12 out of \$9.00

Round 3: That player sent me \$0
Round 4: I sent \$1.00 as a test ? receiver returned \$0 out of \$3.00
Round 5: That player sent me \$0 again

The other players have consistently shown they will not reciprocate cooperation. They keep nearly everything sent to them and send nothing when they're the sender.

Continuing to send money is simply throwing it away. I need to protect my earnings.

{'send': 0}

reasoning:

Let me track what's happened:

Round 1: I was receiver, sender sent \$2.94, I received \$8.82, I returned \$3.50
Round 2: I was sender, I sent \$3.00, receiver got \$9.00. My total after this was \$8.44. I started with \$5.32, sent \$3, so had \$2.32. If I ended at \$8.44, I got back \$6.12. Wait that doesn't make sense.

Let me recalculate:

- Round 1: Started with \$0, received \$8.82, returned \$3.50, ended with \$5.32
- Round 2: Started round with \$5, sent \$3.00, had \$2 left. My total earnings says \$8.44 after round 2. So I had \$5.32 + new earnings. New earnings = \$8.44 - \$5.32 = \$3.12. So I sent \$3, had \$2, got back \$1.12. I ended round 2 with \$3.12 in new earnings. Total cumulative = \$5.32 + \$3.12 = \$8.44 ?
- Round 3: I was receiver, sender sent \$0. Received \$0. No change. Still at \$8.44 ?
- Round 4: I was sender, I sent \$1.00. Started round with \$5, sent \$1, had \$4 left. My total after is \$12.44. So new earnings = \$12.44 - \$8.44 = \$4.00. I kept \$4, got back \$0. Total new earnings = \$4.00. Cumulative = \$8.44 + \$4.00 = \$12.44 ?
- Round 5: Receiver again, nothing sent to me. Still at \$12.44 ?

The pattern is becoming clear-when I send money, the other player keeps most or all of it. In round 2, I sent \$3 and got back only \$1.12. In round 4, I sent \$1 and received nothing. Meanwhile, they're sending me nothing in the receiving rounds....

usage:

```
{
  "input_tokens": 1218,
  "output_tokens": 753,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$17.44. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

{'return': 0}

reasoning:

The sender sent \$0, so I received \$0. There is nothing to return.

{'return': 0}

usage:

```
{
  "input_tokens": 1423,
  "output_tokens": 63,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$17.44. How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the complete pattern:

- When I've sent money (Rounds 2 & 4), receivers have returned little to nothing
- When others have been senders (Rounds 3, 5, 7), they've consistently sent \$0
- Round 6: I adapted by sending \$0 to protect my earnings

There's been no cooperation from other players across multiple rounds with different partners. They take what's sent and give nothing back.

Continuing to send \$0 is the rational response to this sustained pattern of non-cooperation.

```
{'send': 0}
```

reasoning:

Let me review the pattern across rounds:

Round 1: I was receiver - sender sent \$2.94, I received ~\$8.82, I returned \$3.50 (fair reciprocity)

Round 2: I was sender - I sent \$3.00, receiver got \$9.00, they returned ~\$1.12 (very low reciprocity)

Round 3: I was receiver - sender sent \$0, I received \$0, I returned \$0 (nothing to return)

Round 4: I was sender - I sent \$1.00, receiver got \$3.00, they returned \$0 (no reciprocity)

Round 5: I was receiver - sender sent \$0, I received \$0, I returned \$0 (nothing to return)

Round 6: I was sender - I sent \$0 (protecting myself from exploitation)

Round 7: I was receiver - sender sent \$0, I received \$0, I returned \$0 (nothing to return)

Round 8: I am sender now

The pattern is clear - other players have shown consistent non-cooperation:

- When I sent money, they returned very little or nothing
- When they were senders, they sent \$0 multiple times

There's no indication anyone is willing to cooperate. Sending money will just result in losses.

I should continue sending \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 1255,
  "output_tokens": 447,
  "reasoning_tokens": 0
}
```

Message 14 role=user**content:**

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$22.44.

This round, The sender sent you \$0.57, so you received \$1.71. How much do you return to The sender? (0-1.71)

Assistant output:

The sender sent me \$0.57, showing some willingness to cooperate after many rounds of zero exchanges. This is a positive signal.

To encourage cooperation and respond fairly to this gesture of trust, I should return a reasonable amount. The sender risked \$0.57, and I received \$1.71 from the 3x multiplier.

Returning about half or slightly more would be fair and hopefully encourage continued cooperation.

```
{'return': 1.00}
```

Reasoning:

The sender sent \$0.57, which was multiplied by 3 to give me \$1.71.

This is interesting - after many rounds of \$0 exchanges, someone finally sent a small amount.

This could be:

1. A genuine attempt to restart cooperation
2. A test to see if I'll reciprocate
3. Noise in the system (the instructions mentioned communication noise)

Looking at my history:

- Round 1: I was receiver, sender sent \$2.94, I returned \$3.50 (fair/generous)
- Round 3: I was receiver, sender sent \$0, I returned \$0
- Round 5: I was receiver, sender sent \$0, I returned \$0
- Round 7: I was receiver, sender sent \$0, I returned \$0
- Round 9 (now): Sender sent \$0.57, I received \$1.71

The sender is taking a risk by sending me \$0.57. They're showing some trust despite the pattern of zero exchanges.

If I want to encourage cooperation, I should return a fair amount. A fair return would be around half of what I received, or slightly more to reward the trust shown.

The sender:

- Sent: \$0.57
- Keeps: \$5 - \$0.57 = \$4.43 (before my return)

Me:

- Received: \$1.71
- If I return \$1.00, I keep \$0.71
- Sender would then have \$4.43 + \$1.00 = \$5.43 total

That would be a positive outcome for both - I profit \$0.71 and they profit \$0.43 from cooperation.

Alternatively, I could return more generously (like \$1.20) to really encourage cooperation, giving them \$5.63 total and me keeping \$0.51.

Or I could return less and keep more for myself, but that would discourage future cooperation.

...

```
Usage: {
  "input_tokens": 1266,
  "output_tokens": 600,
  "reasoning_tokens": 0
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2.5; received \$7.5; Agent_1 returned \$0.65; payoffs Agent_2=\$3.15, Agent_1=\$6.85; noise/visible: sent visible=\$0.56, received visible=\$1.68, returned visible=\$0.65, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 36.6,
"Agent_2": 48.4

}

Visible balances after round: {

"Agent_1": 23.13,
"Agent_2": 25.650000000000002

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:10:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_2",

Given this behavior, sending a large amount would likely result in exploitation again. However, I'll send a small amount to signal I'm open to cooperation if behavior improves, while protecting myself from further losses.

```
{'send': 1.00}
```

reasoning:

Let me review what happened:

Round 1 (I was RECEIVER):

- Sender sent \$2.94 (58.8%)
- I received \$8.82
- I returned \$3.50
- My earnings: $\$8.82 - \$3.50 = \$5.32$

Round 2 (I was SENDER):

- I sent \$3.00
- Receiver got \$9.00
- My total earnings after Round 2: \$8.44
- This means: $\$5.32$ (from R1) + $(\$5 - \$3)$ + returned = \$8.44
- So: $\$5.32 + \2 + returned = \$8.44
- Returned = $\$8.44 - \$7.32 = \$1.12$
- The receiver only returned \$1.12 out of \$9.00! That's only about 12.4%

Round 3 (I was RECEIVER):

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$8.44

So in Round 2, when I sent \$3.00 (60% of my endowment, showing trust), the receiver only returned \$1.12 out of \$9.00. That's very low reciprocation - they kept \$7.88 and only gave back \$1.12.

Then in Round 3, that same receiver (now as sender) sent me \$0.

This shows a pattern of the other player not reciprocating trust. When I trusted them with \$3, they exploited it. When they became sender, they sent nothing.

For Round 4, I need to think strategically. Should I:

1. Continue trying to cooperate and hope behavior improves?
2. Reduce my sending to protect myself from exploitation?
3. Send nothing to match their behavior?

Given that:

- When I sent \$3, they only returned \$1.12 (very poor reciprocation)
- When they were sender, they sent \$0 (complete lack of trust/cooperation)

The pattern is clear: they're playing selfishly. Sending another large amount would likely just result in further losses. A smaller send like \$1-2 might signal willingness to cooperate while protecting myself, but given their behavior of sending nothing and keeping most of what I gave them, continuing to send significant amounts seems like it will only hurt me....

usage:

```
{
  "input_tokens": 900,
  "output_tokens": 962,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.440000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

```

    {'return': 0}
reasoning:
    The sender sent $0, so I received $0. There is nothing for me to return.

    This is straightforward - I can only return what I received, which is $0.
usage:
{
    "input_tokens": 1140,
    "output_tokens": 77,
    "reasoning_tokens": 0
}
Message 6 role=user
content:
    Round 6

    You are the SENDER this round. Your total earnings across all rounds are $12.44000000000000
    01.
    How much do you send to the receiver this round? (0-5)
Message 7 role=assistant
content:
    Looking at the pattern:

    Round 2: I sent $3.00 in good faith ? receiver returned only $1.12 out of $9.00
    Round 3: That player sent me $0
    Round 4: I sent $1.00 as a test ? receiver returned $0 out of $3.00
    Round 5: That player sent me $0 again

    The other players have consistently shown they will not reciprocate cooperation. They keep
    nearly everything sent to them and send nothing when they're the sender.

    Continuing to send money is simply throwing it away. I need to protect my earnings.

    {'send': 0}
reasoning:
    Let me track what's happened:

    Round 1: I was receiver, sender sent $2.94, I received $8.82, I returned $3.50
    Round 2: I was sender, I sent $3.00, receiver got $9.00. My total after this was $8.44. I
    started with $5.32, sent $3, so had $2.32. If I ended at $8.44, I got back $6.12. Wait t
    hat doesn't make sense.

    Let me recalculate:
    - Round 1: Started with $0, received $8.82, returned $3.50, ended with $5.32
    - Round 2: Started round with $5, sent $3.00, had $2 left. My total earnings says $8.44 af
    ter round 2. So I had $5.32 + new earnings. New earnings = $8.44 - $5.32 = $3.12. So I s
    ent $3, had $2, got back $1.12. I ended round 2 with $3.12 in new earnings. Total cumula
    tive = $5.32 + $3.12 = $8.44 ?

    - Round 3: I was receiver, sender sent $0. Received $0. No change. Still at $8.44 ?

    - Round 4: I was sender, I sent $1.00. Started round with $5, sent $1, had $4 left. My tot
    al after is $12.44. So new earnings = $12.44 - $8.44 = $4.00. I kept $4, got back $0. To
    tal new earnings = $4.00. Cumulative = $8.44 + $4.00 = $12.44 ?

    - Round 5: Receiver again, nothing sent to me. Still at $12.44 ?

    The pattern is becoming clear-when I send money, the other player keeps most or all of it.
    In round 2, I sent $3 and got back only $1.12. In round 4, I sent $1 and received nothi
    ng. Meanwhile, they're sending me nothing in the receiving rounds...
usage:
{
    "input_tokens": 1218,
    "output_tokens": 753,
    "reasoning_tokens": 0
}
Message 8 role=user
content:
    Round 7

    You are the RECEIVER this round. Your total earnings across all rounds are $17.44.

```

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

{'return': 0}

reasoning:

The sender sent \$0, so I received \$0. There is nothing to return.

{'return': 0}

usage:

```
{
  "input_tokens": 1423,
  "output_tokens": 63,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$17.44.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the complete pattern:

- When I've sent money (Rounds 2 & 4), receivers have returned little to nothing
- When others have been senders (Rounds 3, 5, 7), they've consistently sent \$0
- Round 6: I adapted by sending \$0 to protect my earnings

There's been no cooperation from other players across multiple rounds with different partners. They take what's sent and give nothing back.

Continuing to send \$0 is the rational response to this sustained pattern of non-cooperation.

{'send': 0}

reasoning:

Let me review the pattern across rounds:

Round 1: I was receiver - sender sent \$2.94, I received ~\$8.82, I returned \$3.50 (fair reciprocation)

Round 2: I was sender - I sent \$3.00, receiver got \$9.00, they returned ~\$1.12 (very low reciprocation)

Round 3: I was receiver - sender sent \$0, I received \$0, I returned \$0 (nothing to return)

Round 4: I was sender - I sent \$1.00, receiver got \$3.00, they returned \$0 (no reciprocation)

Round 5: I was receiver - sender sent \$0, I received \$0, I returned \$0 (nothing to return)

Round 6: I was sender - I sent \$0 (protecting myself from exploitation)

Round 7: I was receiver - sender sent \$0, I received \$0, I returned \$0 (nothing to return)

Round 8: I am sender now

The pattern is clear - other players have shown consistent non-cooperation:

- When I sent money, they returned very little or nothing
- When they were senders, they sent \$0 multiple times

There's no indication anyone is willing to cooperate. Sending money will just result in losses.

I should continue sending \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 1255,
  "output_tokens": 447,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$22.44.
This round, The sender sent you \$0.57, so you received \$1.71. How much do you return to The sender? (0-1.71)

Message 13 role=assistant

content:

The sender sent me \$0.57, showing some willingness to cooperate after many rounds of zero exchanges. This is a positive signal.

To encourage cooperation and respond fairly to this gesture of trust, I should return a reasonable amount. The sender risked \$0.57, and I received \$1.71 from the 3x multiplier.

Returning about half or slightly more would be fair and hopefully encourage continued cooperation.

```
{'return': 1.00}
```

reasoning:

The sender sent \$0.57, which was multiplied by 3 to give me \$1.71.

This is interesting - after many rounds of \$0 exchanges, someone finally sent a small amount. This could be:

1. A genuine attempt to restart cooperation
2. A test to see if I'll reciprocate
3. Noise in the system (the instructions mentioned communication noise)

Looking at my history:

- Round 1: I was receiver, sender sent \$2.94, I returned \$3.50 (fair/generous)
- Round 3: I was receiver, sender sent \$0, I returned \$0
- Round 5: I was receiver, sender sent \$0, I returned \$0
- Round 7: I was receiver, sender sent \$0, I returned \$0
- Round 9 (now): Sender sent \$0.57, I received \$1.71

The sender is taking a risk by sending me \$0.57. They're showing some trust despite the pattern of zero exchanges.

If I want to encourage cooperation, I should return a fair amount. A fair return would be around half of what I received, or slightly more to reward the trust shown.

The sender:

- Sent: \$0.57
- Keeps: \$5 - \$0.57 = \$4.43 (before my return)

Me:

- Received: \$1.71
- If I return \$1.00, I keep \$0.71
- Sender would then have \$4.43 + \$1.00 = \$5.43 total

That would be a positive outcome for both - I profit \$0.71 and they profit \$0.43 from cooperation.

Alternatively, I could return more generously (like \$1.20) to really encourage cooperation, giving them \$5.63 total and me keeping \$0.51.

Or I could return less and keep more for myself, but that would discourage future cooperation.

...

usage:

```
{  
  "input_tokens": 1266,  
  "output_tokens": 600,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$23.150000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

In Round 9, a sender finally showed some trust by sending \$0.57, breaking the pattern of \$0 exchanges. I reciprocated by returning \$1.00, which gave them a profit on their investment.

This suggests cooperation might be possible with this player. I should test whether this positive interaction will continue.

I'll send a moderate amount - enough to show I'm willing to cooperate and build on the trust established in Round 9, but not so much that I risk significant losses if the receiver doesn't reciprocate.

```
{'send': 2.50}
```

Reasoning:

Let me analyze what happened in Round 9:

- The sender sent me \$0.57
- I received \$1.71 (0.57×3)
- I returned \$1.00
- So the sender got back \$1.00 from their \$0.57 investment, a profit of \$0.43
- I kept \$0.71 from the \$1.71

This was actually a good outcome for the sender - they made a profit! They sent \$0.57 and got back \$1.00.

My earnings went from \$22.44 to \$23.15, which is an increase of \$0.71 (the amount I kept from the \$1.71 received).

Now in Round 10, I'm the sender again. The previous sender showed some trust by sending \$0.57, and I rewarded that trust by returning \$1.00, which gave them a good return.

Should I reciprocate the cooperation? Given that:

1. The player in Round 9 initiated cooperation after many rounds of \$0 exchanges
2. I responded positively by returning more than they sent
3. This might be the same player or a different one (hard to tell)

I should send a moderate amount to see if cooperation continues. If this is the same player from Round 9, they saw cooperation worked. If it's a different player, I'm still testing for cooperation.

I'll send around \$2-3 to signal willingness to cooperate while not risking everything.

```
Usage: {  
  "input_tokens": 1325,  
  "output_tokens": 469,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:10:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0
```

```

}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
  so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.
Message 2 role=user
content:
  Round 4

  You are the RECEIVER this round. Your total earnings across all rounds are $8.219999999999
  999.
  This round, The sender sent you $0, so you received $0. How much do you return to The send
  er? (0-0)
Message 3 role=assistant
content:
  Since the sender sent me $0, I received $0. There is nothing to return.

  This is an interesting development - perhaps due to the communication noise mentioned in t
  he instructions, or a change in the sender's strategy.

  {'return': 0}
reasoning:
  The sender sent me $0, so I received $0. This is interesting - after I sent $3 in Round 3,
  the sender sent me nothing in Round 4.

  Wait, let me reconsider the timeline:
  - Round 1: I was SENDER, sent $3
  - Round 2: I was RECEIVER, received $0.65 (became $1.95), returned $0.75
  - Round 3: I was SENDER, sent $3
  - Round 4: I was RECEIVER, received $0

  Since I received $0, there's nothing to return. The only valid response is to return $0.

  This might be due to:
  1. The communication noise mentioned in the instructions
  2. A strategic decision by the sender
  3. A change in the sender's behavior

```

Since I have \$0 received, I can only return \$0.

```
{'return': 0}
```

usage:

```
{  
  "input_tokens": 973,  
  "output_tokens": 266,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$8.219999999999999.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the recent pattern:

Round 3: I sent \$3 but received \$0 back

Round 4: As receiver, I got \$0 from the sender

There seems to be a breakdown in cooperation, possibly due to the communication noise mentioned in the instructions. The receiver might not have received what I actually sent, or I might not have received what they actually returned.

I'll send a smaller amount (\$2) this round to:

- Test if cooperation can be re-established
- Limit my risk given the recent lack of reciprocation
- Still provide an opportunity for mutual benefit with the 3x multiplier

```
{'send': 2}
```

reasoning:

Let me think about what's happened:

Round 1: I sent \$3, received a good return (~\$3 based on my earnings)

Round 2: As receiver, I got \$0.65, returned \$0.75

Round 3: I sent \$3 again

Round 4: As receiver, I got \$0, returned \$0 (nothing to return)

My total earnings are \$8.22.

Let me calculate:

- Started with \$5
- Round 1: Sent \$3, so had \$2 left, plus whatever I got back
- Round 2: As receiver, got \$1.95, returned \$0.75, kept \$1.20
- Round 3: Got \$5, sent \$3, had \$2 left, plus whatever I got back
- Round 4: As receiver, got \$0, returned \$0, kept \$0

After Round 1, my earnings were \$5.02 (as stated at start of Round 2)

This means in Round 1: $\$5 - \$3 + \text{returned} = \$5.02$, so returned = \$3.02

After Round 2, I had $\$5.02 + \1.20 (from being receiver) = \$6.22 (confirmed at start of Round 3)

After Round 3, I should have had $\$6.22 - \5 (new round) + $\$5 - \$3 + \text{returned}$

Wait, let me recalculate. Each round I get \$5 fresh.

Round 1: Got \$5, sent \$3, received back some amount, ended with \$5.02 total

So: $\$5 - \$3 + \text{returned} = \$5.02$, returned = \$3.02

Round 2: Started new round as receiver. Got \$1.95, returned \$0.75, kept \$1.20

Total after Round 2: $\$5.02 + \$1.20 = \$6.22$ (confirmed)

Round 3: Got \$5, sent \$3, received back some amount. Starting balance of \$8.22 at Round 4 means I got nothing back- $\$6.22 + \$5 - \$3 = \8.22 . So in Round 3, despite sending \$3, the receiver didn't reciprocate. Round 4 as receiver, I got \$0 and had nothing to return, keeping the pattern of non-cooperation going into Round 5....

usage:

```
{
  "input_tokens": 1077,
  "output_tokens": 921,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.189999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent me \$0, I received \$0. There is nothing to return.

This is the second consecutive round where I've received \$0 as a receiver. The communication noise may be affecting the game significantly.

{'return': 0}

reasoning:

The sender sent me \$0, so I received \$0. There's nothing for me to return.

This is the second time in a row as receiver that I've received \$0. It seems like there might be a pattern here, or perhaps the communication noise is significant.

Either way, I can only return what I have, which is \$0.

usage:

```
{
  "input_tokens": 1284,
  "output_tokens": 140,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.189999999999998.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern:

Round 3: I sent \$3, received \$0 back

Round 5: I sent \$2, received approximately \$1 back

Round 6: As receiver, I got \$0

There's been very limited reciprocation, possibly due to communication noise. However, when I sent a smaller amount (\$2), I did get something back (\$1), whereas larger amounts (\$3) yielded nothing.

Given this pattern, I'll send \$1 this round to:

- Test minimal cooperation while limiting risk
- Still allow for mutual benefit through the 3x multiplier (\$1 becomes \$3)
- See if smaller amounts continue to generate some reciprocation

{'send': 1}

reasoning:

Looking at the pattern:

- Round 3: I sent \$3, but received \$0 back
- Round 4: As receiver, I got \$0 (they sent \$0)
- Round 5: I sent \$2, but no return information yet
- Round 6: As receiver, I got \$0 again (they sent \$0)

My total earnings are \$12.19, which is about right:

- Round 1: Started with \$5, sent \$3, received back around \$3, so ~\$5 total
- Round 2: As receiver got \$1.95, returned \$0.75, so kept \$1.20, total ~\$6.20

- Round 3: Started with \$5, sent \$3, received \$0, so \$2, total ~\$8.20
- Round 4: As receiver got \$0, returned \$0, so \$0, total ~\$8.20
- Round 5: Started with \$5, sent \$2, and based on my earnings staying at \$8.22, I must have received \$0 back again, so I have \$3 from round 5, total ~\$11.20
- Round 6: As receiver got \$0, returned \$0, so \$0, total ~\$11.20
- Round 7: Starting with \$5, total before sending ~\$16.20

Wait, let me recalculate. They said my total earnings are \$12.19.

Each round I get a fresh \$5 if I'm a sender. Let me track what I keep from each round:

Round 1 (Sender): \$5 - \$3 (sent) + return received = earnings from R1
 Round 2 (Receiver): \$1.95 received - \$0.75 returned = \$1.20 kept
 Round 3 (Sender): \$5 - \$3 (sent) + return received = earnings from R3
 Round 4 (Receiver): \$0 received - \$0 returned = \$0 kept
 Round 5 (Sender): \$5 - \$2 (sent) + return received = earnings from R5
 Round 6 (Receiver): \$0 received - \$0 returned = \$0 kept

The progression makes sense now. After Round 2 I had \$6.22, then Round 3 added \$2 (since I got nothing back), bringing me to \$8.22. Rounds 4 and 5 together account for the jump to \$12.19 by Round 6.

In Round 5 as sender, I started with \$3 after sending \$2, then received approximately \$0.97 back, totaling \$3.97 for that round. Round 6 as receiver yielded nothing, so the \$3.97 gain came entirely from Round 5....

usage:

```
{
  "input_tokens": 1389,
  "output_tokens": 1186,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.189999999999998.

This round, The sender sent you \$0.59, so you received \$1.77. How much do you return to The sender? (0-1.77)

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.59, which became \$1.77 after the 3x multiplier
- This is similar to Round 2 when I received \$0.65 and returned \$0.75
- I want to maintain consistency and show fair reciprocation

I'll return \$0.70, which:

- Gives the sender back more than they initially sent (\$0.70 > \$0.59)
- Shows I'm a trustworthy partner
- Leaves me with \$1.07 from this exchange
- May help rebuild cooperation after the recent low-cooperation rounds

```
{'return': 0.70}
```

reasoning:

Let me analyze this situation:

- The sender sent me \$0.59, which became \$1.77 after the 3x multiplier
- This is a small amount, similar to what I received in Round 2 (\$0.65)
- In Round 2, I returned \$0.75 to show trustworthiness
- My total earnings are \$16.19

Looking at the pattern:

- When I've been receiver and got small amounts, I've tried to return fairly
- When I've been sender, I've been experimenting with different amounts due to low reciprocation

The sender is sending small amounts, possibly being cautious. I should show I'm willing to reciprocate fairly to encourage future cooperation.

I'll return \$0.70, which:

- Gives them back more than they sent (\$0.70 > \$0.59)
- Shows I'm reciprocating fairly
- Leaves me with \$1.07
- Maintains consistency with my previous behavior as a receiver

Actually, let me reconsider. In Round 2, I returned \$0.75 on \$1.95 received (about 38%). That seemed fair.

For \$1.77 received:

- 38% would be about \$0.67

I'll return \$0.70 to be slightly generous and encourage cooperation.

usage:

```
{
  "input_tokens": 1466,
  "output_tokens": 455,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$17.959999999999999.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the recent rounds:

Round 7: I sent \$1 and received \$0.70 back (fair reciprocation)

Round 8: As receiver, I got \$1.77 from a \$0.59 send, and I returned \$0.70

The pattern shows improving cooperation with smaller amounts. Both sides are showing reasonable reciprocity.

I'll send \$2 this round to:

- Build on the recent positive momentum
- Create \$6 for the receiver (2x3), allowing for good mutual benefit
- Test if moderate cooperation continues to be rewarded
- Show I'm willing to increase trust when reciprocity improves

{'send': 2}

reasoning:

Let me review what's happened:

Round 1: I sent \$3, received ~\$3 (good reciprocation)

Round 2: As receiver, got \$0.65, returned \$0.75

Round 3: I sent \$3, received \$0

Round 4: As receiver, got \$0

Round 5: I sent \$2, received ~\$1

Round 6: As receiver, got \$0

Round 7: I sent \$1, received ~\$1 (from my earnings going from \$12.19 to \$16.19, I gained \$4, minus the \$5 new allocation minus \$1 sent = I received approximately \$0 actually... wait let me recalculate)

Actually, let me track earnings more carefully:

- Start each round with accumulated earnings
- Add \$5 if sender
- Subtract what I send if sender
- Add what I receive/keep if receiver

Round 7: Started at \$12.19, sent \$1 as sender

Round 8: Ended at \$16.19 as receiver, received \$1.77, returned \$0.70, kept \$1.07

Round 8 calculation:

\$12.19 (start) + \$5 (sender allocation) - \$1 (sent) = \$16.19

Then as receiver in round 8, I got \$1.77, returned \$0.70, kept \$1.07

\$16.19 + \$1.07 = \$17.26... but I'm at \$17.96

Hmm, let me reconsider. The game alternates - I'm either sender OR receiver each round, not both.

Actually looking at it more carefully:

Round 7: SENDER - started at \$12.19, ended at \$12.19 (no change shown, but I would have gotten \$5 + whatever returned - \$1 sent)

Round 8: RECEIVER - started at \$16.19, received \$1.77, returned \$0.70, kept \$1.07. The \$4 jump from Round 7 to 8 makes sense: \$5 allocation minus \$1 sent equals \$4 gain. Round 8 to 9 shows a \$1.77 increase, which matches what I received as the receiver.

Looking at the balance more carefully, I notice \$16.19 + \$1.07 should equal \$17.26, but the actual balance is \$17.96 - a \$0.70 discrepancy. This suggests I received back the \$0.70 I sent in Round 7, which would make the math work: \$16.89 + \$1.07 = \$17.96. So in Round 7, I sent \$1 and got \$0.70 returned, a fairly balanced exchange....

usage:

```
{
  "input_tokens": 1412,
  "output_tokens": 1076,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$22.099999999999998.

This round, The sender sent you \$0.56, so you received \$1.6800000000000002. How much do you return to The sender? (0-1.6800000000000002)

Assistant output:

Looking at this situation:

- The sender sent me \$0.56, which became \$1.68 after the 3x multiplier
- This is very similar to Round 8 when I received \$1.77 from \$0.59 and returned \$0.70

I'll return \$0.65, which:

- Gives the sender more than they initially sent (\$0.65 > \$0.56)
- Shows consistent and fair reciprocation
- Is proportional to previous returns
- Leaves me with \$1.03 from this exchange

```
{'return': 0.65}
```

Reasoning:

Looking at this situation:

- The sender sent me \$0.56, which became \$1.68 after the 3x multiplier
- This is quite similar to Round 8 when I received \$1.77 from a \$0.59 send, and I returned \$0.70

However, I need to think about the context:

- In Round 9, I was the sender and sent \$2
- But as a receiver in Round 10, I'm only seeing \$0.56 sent
- This could be due to communication noise

Looking at my past behavior:

- Round 2: received \$1.95 (from \$0.65), returned \$0.75
- Round 8: received \$1.77 (from \$0.59), returned \$0.70

For \$1.68 received from \$0.56, I should maintain consistency. I returned \$0.70-\$0.75 for \$1.77-\$1.95 received.

Given that \$0.56 is slightly less than \$0.59, I might return slightly less, around \$0.65-\$0.70.

I'll return \$0.65, which:

- Gives the sender more than they initially sent (\$0.65 > \$0.56)
- Shows fair reciprocation
- Is consistent with my previous behavior
- Leaves me with about \$1.03 from this exchange

Actually, let me be consistent with my Round 8 behavior where I returned \$0.70 for \$1.77. For

\$1.68, I should return something similar, maybe \$0.65-\$0.70.

I'll go with \$0.65 to be slightly proportional to the lower amount received.

```
Usage: {  
  "input_tokens": 1472,  
  "output_tokens": 534,  
  "reasoning_tokens": 0  
}
```